Cyber-resilience in Smart Water Networks: A Systematic Review and the TRITON Framework for Secure AI-enabled Water Infrastructure

Isaac David Torres Paredes[1*], Luis Enrique Sánchez Crespo[1], Antonio Santos-Olmo[1], David G. Rosado[1], Eduardo Fernández-Medina[1]

[1*]GSyA Research Group, Department of Information Technology and Systems, University of Castilla-La Mancha, Ciudad Real, Spain.

*Corresponding author(s). E-mail(s): IsaacDavid.Torres1@alu.uclm.es;
Contributing authors: luise.sanchez@uclm.es;
antonio.santosolmo@uclm.es; david.grosado@uclm.es;
eduardo.fdezmedina@uclm.es;

Abstract

The convergence of IoT sensors, SCADA systems, and cloud-based analytics in drinking water distribution networks has expanded the cyber-attack surface of these critical infrastructures, exposing them to threats ranging from false data injection to denial-of-service disruptions—risks that are amplified in developing regions where governance capacity and security investment remain limited. This paper presents a systematic review, following the PRISMA 2020 protocol, of 40 primary studies drawn from 1 580 records across six academic sources (2015–2026), examining how cybersecurity, sustainability, and artificial intelligence are addressed in smart water network research.

The cross-domain analysis reveals that the three domains are treated predominantly in isolation: 55% of the reviewed studies (22 of 40) address a single thematic axis, 45% (18 of 40) incorporate elements from two axes, and none of the studies included in the final corpus was coded as providing substantive coverage of all three simultaneously. Only 13 of the 467 cybersecurity-related records retrieved met the inclusion criteria, indicating limited coverage of security-focused research specific to the water sector within the retrieved corpus. Recurring vulnerabilities include unpatched SCADA firmware, absence of network segmentation between IT and OT zones, and lack of incident response protocols tailored to water operations.

To address this integration deficit, we propose the TRITON* framework (Technology Resilience, cybersecurity and Intelligence for Transformative Optimization of smart water Networks), a layered architecture that organises ten cybersecurity controls (CI-1 to CI-10), eight sustainability indicators (SO-1 to SO-8), and eight AI capabilities (IA-1 to IA-8) into a phased deployment model designed for resource-constrained water utilities. The detailed technical specification of each layer and its application to a real-world campus network are presented in a companion paper.

Keywords: Cybersecurity, Critical Infrastructure, Smart Water Networks, SCADA Security, Artificial Intelligence, Sustainability, TRITON

1 Introduction

In recent decades, the efficient and sustainable management of water resources has become a global priority Woldt and Dahab (2006); Velayudhan et al. (2022); Gupta et al. (2020). Urban population growth, increasing water scarcity, ageing infrastructure, and more stringent environmental regulations have driven water utilities to explore smart water network solutions Ramos et al. (2019); di Nardo et al. (2021); Cahn (2014); Fernández-Ahumada et al. (2019). Such solutions integrate emerging technologies—sensors, wireless networks, predictive modelling, and information systems—to optimise the efficiency, reliability, and sustainability of water systems Cuguero-Escofet and Puig (2023); Osorio et al. (2018); Soto-García et al. (2013). Smart water networks have thus emerged as a promising alternative for substantially improving supply and treatment system management Garrido-Baserba et al. (2020); Tuptuk et al. (2021); de Paula et al. (2019).

Operationally, these networks rely on information and communication technologies (ICTs) for monitoring, control, and resource management Sun et al. (2018). Machine learning enables short- and long-term prediction of water consumption; sensor and wireless network integration facilitates real-time condition monitoring, contributing to early detection of leaks and anomalies Harshadeep and Young (2020). Big Data supports the modelling and analysis of large datasets, thereby optimising operation and maintenance Serra et al. (2014); Hariveena et al. (2020). The Internet of Things (IoT) stands out as one of the most widely adopted technologies for intelligent monitoring and control of distribution systems Jagtap et al. (2021); Taormina et al. (2017); Bour et al. (2023).

Cybersecurity, however, represents a critical challenge in smart water network deployment Taormina et al. (2018). The convergence of legacy SCADA protocols with IP-connected IoT devices widens the attack surface, exposing water systems to false data injection, ransomware, and denial-of-service threats that can compromise both public health and supply continuity Clark et al. (2017); Hassanzadeh et al. (2020);

*The acronym TRITON should not be confused with the TRITON/TRISIS malware (MITRE ATT&CK ICS S1009) that targets Schneider Electric Triconex safety controllers. In this work, TRITON refers exclusively to the proposed defensive framework.

Pérez-Padillo et al. (2020). Analogous vulnerabilities have been documented in smart grid infrastructures, where SCADA-targeted attacks exploit unpatched firmware and flat OT/IT topologies (Achaal et al. 2024). Process-aware stealthy attacks—which manipulate sensor readings within plausible operating ranges—pose a particular risk to water treatment systems, as demonstrated by recent advances in ellipsoid-boundary detection models (Maurya et al. 2023). The widely reported 2021 Oldsmar incident, although its precise attribution remains disputed, illustrates the potential safety consequences of insecure remote access and weak operational safeguards in water treatment environments. The adoption of Zero Trust principles in IoT environments (Liu et al. 2024) and ensemble-learning-based anomaly detection (Lai et al. 2024) offer promising countermeasures, yet their application to the water sector remains largely unexplored. As dependence on ICTs grows, so does the urgency of securing network integrity Batarseh et al. (2023); Saad et al. (2020); Skiba (2020); Ali (2021). A structured approach to cybersecurity that also accounts for sustainability and AI-driven efficiency is therefore needed.

Within this context, the aim of this work is to conduct a systematic literature review of ICTs applied to smart and sustainable water networks, identifying research gaps and current trends in order to propose a framework that addresses cybersecurity and efficiency challenges. The contributions of this paper are as follows:

1. A systematic literature review that identifies and categorises 40 unique primary studies addressing the application of cybersecurity, sustainability, and artificial intelligence in smart water networks, classifying them into processes, frameworks, and models.
2. A cross-axis analysis that examines the extent to which existing proposals address one, two, or all three thematic axes simultaneously, thereby revealing critical integration gaps in the current body of knowledge.
3. The TRITON framework (Technology Resilience, cybersecurity and Intelligence for Transformative Optimization of smart water Networks), a layered conceptual architecture that integrates cybersecurity, sustainability, and artificial intelligence components to address the shortcomings identified in the review.

Several systematic reviews have examined individual facets of smart water network management in recent years, yet none has attempted the cross-domain integration pursued here. Habib et al. (2026) surveyed detection methods for anomalies in water distribution networks, concentrating on machine-learning classifiers and their operational constraints; their scope, however, excluded sustainability indicators and governance-level cybersecurity frameworks. Galarza Yallico and Santos López (2024) compiled intrusion detection techniques tailored to SCADA-based water systems and offered a granular taxonomy of attack vectors, but that review did not consider resource optimisation or environmental performance. On the sustainability side, Quintana et al. (2025) catalogued smart water technologies along a technological-readiness axis, documenting sensor deployments, digital twin pilots, and data analytics platforms without proposing a governance architecture that integrates security considerations. Ghorbani Bam et al. (2025) demonstrated the value of digital twin platforms for

hydraulic simulation, whilst their contribution remained confined to the operational modelling layer and did not extend to cybersecurity or policy alignment.

A recurrent observation across these reviews is that each concentrates on a single thematic domain—be it cybersecurity, sustainability, or artificial intelligence—without examining the interactions among them. The present work addresses this fragmentation through a cross-domain systematic review that simultaneously maps proposals from all three axes, classifying each contribution according to its primary output type (process, framework, or model) and subsequently identifying where intersections and gaps occur. The TRITON framework, introduced in Section 5, is proposed to address these gaps by organising cybersecurity controls, sustainability metrics, and AI-driven analytics as interdependent modules rather than isolated silos. Table 17 provides a structured comparison between TRITON and the frameworks cited above, clarifying the specific dimensions in which each existing proposal falls short of a cross-domain treatment. The detailed technical specification of each TRITON layer—including algorithms, protocols, and standards alignment—is presented in a companion paper.

The remainder of the paper is organised as follows. Section 2 presents the review methodology. Section 2.3 details the selection criteria and execution process. Section 3 describes the structured data collection across three thematic axes—Cybersecurity, Sustainability, and Artificial Intelligence—while Section 4 provides an interpretation of the findings. Section 5 introduces the TRITON framework, and Section 6 presents conclusions and future research directions.

2 Review Method

The systematic review was guided by the following research question: *What processes, frameworks, and models have been proposed in the scientific literature to address cybersecurity, sustainability, and artificial intelligence in the context of smart water networks, and to what extent do existing proposals integrate these three domains?*

This question was formulated to determine the current state of research and to identify gaps in the combined treatment of these thematic axes. On this basis, articles published over the past ten years in relevant journals were reviewed, prioritising those addressing emerging technologies, cybersecurity challenges, and sustainability strategies in water systems management. No formal external registration of the review protocol was performed. The protocol was documented internally prior to the search execution and is available from the corresponding author upon request.

2.1 Selection Sources

The search for primary studies was carried out using electronic databases—Scopus, Web of Science (WoS), ScienceDirect, SpringerLink, IEEE Xplore, and Google Scholar—accessed both online and through the digital library of the Escuela Superior Politécnica de Chimborazo.[1] Scopus and WoS were selected as the principal indexing services owing to their comprehensive coverage of engineering and computer science venues and their support for citation-based quality indicators. IEEE

[1] https://elibro-net.proxy.espoch.edu.ec/es/lc/espoch/inicio

Table 1 Search Strings by Topic

Thematic Axis	Search Strings
Cybersecurity	"Smart water systems" AND "cybersecurity"; "IoT in water" AND "cybersecurity risks"; "Water management" AND "cyber threats"; "Critical infrastructure" AND "water systems" AND "cybersecurity"; "Smart water grids" AND "data protection"
Sustainability	"Smart water systems" AND "sustainability"; "Water resource management" AND "smart technologies"; "Sustainable water solutions" AND "IoT"; "Smart cities" AND "water sustainability"; "Water conservation" AND "smart technologies"
Artificial Intelligence	"Smart water systems" AND "artificial intelligence"; "AI in water management"; "Machine learning" AND "smart water networks"; "Predictive analytics" AND "water quality"; "AI-driven water monitoring"

Xplore was included to capture conference proceedings and standards-adjacent publications that are prevalent in the cybersecurity and IoT domains. ScienceDirect and SpringerLink were consulted as publisher-specific platforms to retrieve full texts that might not be immediately accessible through the indexing services. Google Scholar served as a supplementary source, particularly useful for identifying grey literature and preprints that have not yet been indexed elsewhere. Manual methods—including the review of journal tables of contents, conference proceedings, and expert-recommended publications—complemented the electronic search to reduce the risk of retrieval bias.

2.2 Source Selection

Selected keywords were combined using Boolean operators "AND" and "OR" to construct thematic search strings. Table 1 presents the complete list of search strings, organised by thematic axis.

All queries were executed against the *title*, *abstract*, and *keywords* fields in Scopus and Web of Science. In IEEE Xplore, the "command search" mode was employed with the same field combination. Google Scholar searches were conducted on the full text owing to the platform's limited field-specific operators. The initial searches were executed between November 2024 and January 2025; a supplementary update was performed in February 2026 to capture the most recent publications.

The study selection procedure was executed in two sequential stages, with documents restricted to the period 2015–2026. One exception was made for Mutchek and Williams Mutchek and Williams (2014), published in 2014, which was retained as a seminal contribution that established the multi-criteria sustainability assessment framework for water systems subsequently adopted by several studies within the corpus; its exclusion would have removed a key conceptual antecedent referenced by later works. In the first stage, the inclusion criterion was applied to the initial pool of retrieved records. Each document's title, keywords, and abstract were examined

to assess topical alignment with at least one of the three thematic axes. Records that made no reference to smart water networks, cybersecurity in water infrastructure, sustainability in water management, or AI-based solutions for water systems were discarded at this point. Grey literature, editorials, and conference abstracts lacking full-text availability were likewise excluded, as were publications not written in English.

In the second stage, the exclusion criterion was applied to the subset of potentially relevant studies. This entailed a detailed reading of the abstract and conclusions of each candidate and, where necessary, a closer examination of the methodology and results sections. Studies were excluded if they: (i) addressed water management without any technological or digital dimension; (ii) focused on technologies unrelated to distribution or treatment networks (e.g. agricultural irrigation sensors with no network component); or (iii) duplicated contributions already captured from another database. When the same study appeared in multiple databases, the version indexed in Scopus or Web of Science was retained to ensure bibliometric traceability. Two researchers independently performed the screening at both stages; disagreements were resolved through discussion with a third researcher. The overall percentage agreement across both stages exceeded 91%, corresponding to a Cohen's $\kappa \approx 0.80$—classified as *substantial agreement* on the Landis and Koch scale. The review protocol follows the PRISMA 2020 guidelines Page et al. (2021); a flow diagram summarising the screening process is provided in Fig. 1.

2.3 Selection Execution

The query execution yielded 467 studies for Cybersecurity, 506 for Sustainability, and 607 for Artificial Intelligence—a combined initial pool of 1 580 records. After removing duplicates across databases, 1 247 unique records remained. The first-stage screening (title, keywords, and abstract) reduced this set to 127 candidate studies. The second-stage screening (detailed reading of abstracts, conclusions, and, where necessary, full-text examination) further narrowed the corpus to 40 unique primary studies: 13 addressing Cybersecurity, 14 Sustainability, and 14 Artificial Intelligence, with one study (Joseph et al. (2022)) contributing to both Sustainability and AI. Within descriptive Tables 2 and 4, Bosco et al. (2022) and Rojek and Studzinski (2019) each receive dual entries because they contribute to two sub-categories (Process and Framework) within their respective axes; all quantitative analyses are computed over the 40 unique publications. The overall inclusion rate of 3.4% reflects the stringency of the criteria and the specificity of the research question. The complete screening process is illustrated in Fig. 1.

To structure the results, selected studies were grouped according to the three fundamental axes and further sub-classified by primary output type:

- **Process**: A set of planned activities arranged in successive phases with the aim of achieving a specific goal.
- **Framework**: A layered structure designed to support or guide the construction of a risk management scheme, encompassing a set of functions within the system along with the relationships between them.

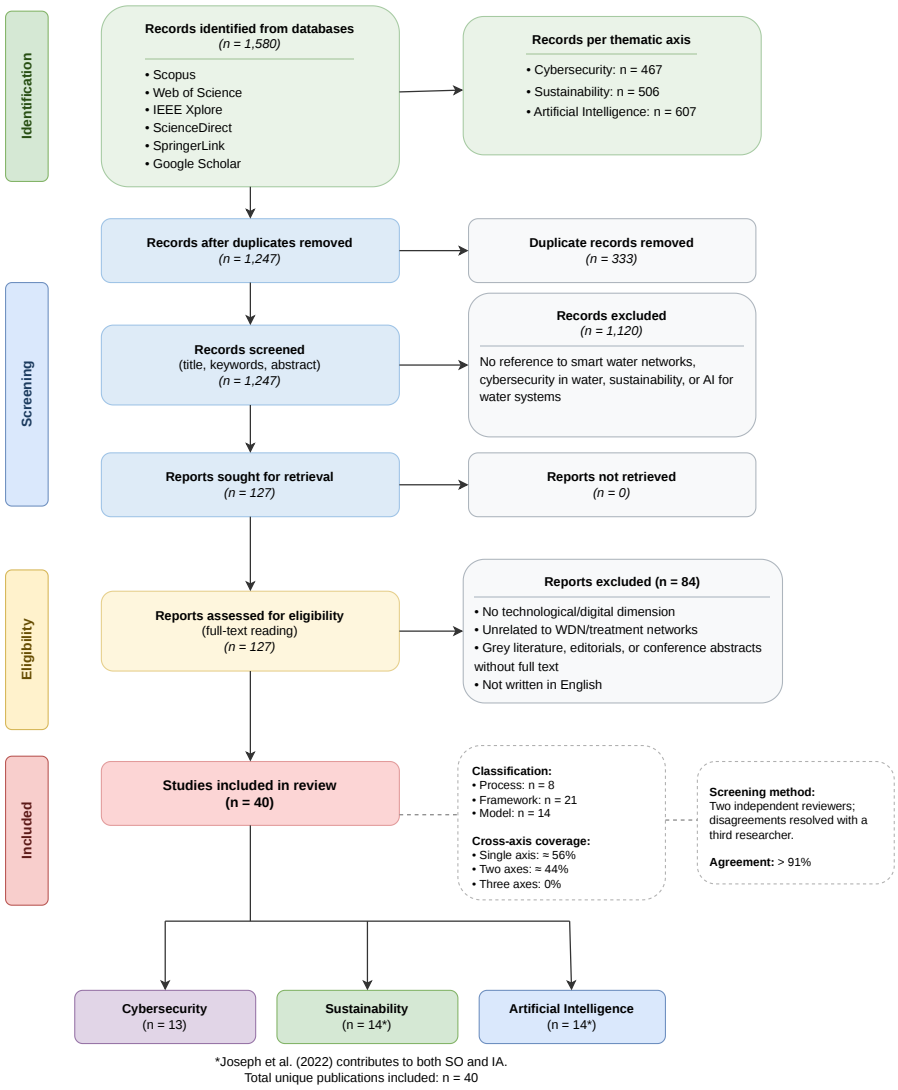


Fig. 1 Search Results and Information Filtering

- **Model**: A specific artefact that provides a representation of a complex system in order to facilitate its understanding.

3 Data Collection

The information gathered from the selected studies was required to include processes, frameworks, or models applied to cybersecurity, sustainability, or artificial intelligence within smart and sustainable water networks. Particular emphasis was placed on identifying approaches that optimise system management, enhance efficiency, and ensure security. For each primary study, a structured data extraction form was completed, recording: (i) bibliographic identification (authors, year, venue, and citation count at the time of extraction); (ii) the methodology or model employed; (iii) the principal results and claimed contributions; (iv) the thematic axes addressed (one, two, or all three); (v) the standards or regulations referenced; and (vi) the stated limitations and open challenges. The completed forms were then cross-tabulated to identify recurring patterns within and across axes. For the comparative tables (Tables 5–7), each study was coded against the domain-specific characteristics using a ternary scale: Y (the study explicitly addresses the characteristic), P (partial or indirect coverage), and N (no coverage). Coding was performed independently by two researchers; where their assessments diverged, the full text of the study was re-examined jointly until consensus was reached. The coding rules were as follows: Y required an explicit mention of the characteristic in the study's methodology, results, or discussion; P was assigned when the characteristic was referenced but not empirically addressed or when the coverage was tangential; N was recorded when no reference to the characteristic could be identified.

Generative AI tools (Claude by Anthropic and Gemini by Google) were used exclusively for linguistic revision, structural formatting, and style adaptation to the journal template; they were not employed for search execution, screening, study selection, or data coding.

To provide a concise and structured synthesis, the detailed findings have been distilled into three summary tables (Tables 2, 3, and 4), each accompanied by a synthesis paragraph and a discussion of representative contributions. The following subsections present the structured synthesis for each thematic axis.

3.1 Cybersecurity

The 13 studies reviewed under the Cybersecurity axis address the protection of water distribution and treatment networks against digital threats. Although several incorporate adjacent technologies—digital twins, Big Data, IoT—the analytical core consistently lies in cybersecurity principles, methodologies, or solutions. The studies were sub-classified into three processes, seven frameworks, and four models.

Table 2 summarises the key contribution and limitations of each reviewed study.

Across the 13 cybersecurity studies, several patterns emerge. Frameworks dominate the landscape, with half of the reviewed contributions proposing structured guidelines or reference architectures—frequently aligned with established standards such as NIST, ISO 27001, and ISO 31000. A recurring gap is the lack of empirical validation: most proposals rely on theoretical analysis or simulation rather than field deployment, and few address the specific constraints of rural or resource-limited utilities. Intrusion detection, SCADA hardening, and risk assessment are the predominant methodological

Table 2 Summary of Cybersecurity studies. Studies marked with † contribute to two sub-categories (Process and Framework) and appear twice; the count of 14 rows corresponds to 13 unique publications.

#	Reference	Type	Key Contribution	Limitations
1	Batarseh et al. (2023)	Process	AI–cybersecurity testbed (ACWA) simulating digital twins for water distribution	No comparison with testbeds capturing chemical composition; limited standardisation for replication
2	Alghassab (2021)	Process	AHP–TOPSIS multi-criteria method for evaluating cyber vulnerabilities in critical infrastructure	Lacks real-world case study validation; limited implementation guidance
3	Bosco et al. (2022)†	Process	Dynamic hydraulic model with real-time simulation and cybersecurity law compliance	Does not address long-term costs or challenges in rural/less-developed areas
4	Bello et al. (2022)	Framework	PRISMA-based systematic review of cybersecurity challenges in Australian water infrastructure	94% of studies from academia; lacks industry–government collaboration analysis
5	Tuptuk et al. (2021)	Framework	Systematic classification of cybersecurity publications for water systems, identifying research trends	Focus limited to potable water; no comprehensive security framework proposed
6	Clark and Panguluri (2018)	Framework	NIST-aligned practical measures (unidirectional gateways, SCADA hardening) for water utilities	Absence of unified federal policy; uneven preparedness across utility sizes
7	Kumar et al. (2019)	Framework	Comparative analysis of CTI platforms; evaluates TAXII, CybOX, and ontology-based threat detection	No focus on non-IT critical infrastructure sectors; trust issues in sharing communities unaddressed
8	Ali (2021)	Framework	Cyclical cyber risk management approach (risk, security, trust, evaluation) aligned with FISMA/NIST	Lacks detailed real-world implementation analysis; omits emerging technology implications for DCS
9	Muttaqin and Kalamullah Ramli (2023)	Framework	Security framework integrating C2M2, ISO 27019, and Indonesian regulations for water sector IT/OT	No prior water-specific frameworks for validation in Indonesian context
10	Bosco et al. (2022)†	Framework	STOP-IT platform integrating 28 solutions for cyber–physical risk management (ISO 31000)	Digital literacy gaps hinder adoption; platform complexity may be a barrier
11	Lin et al. (2024)	Model	Mathematical framework quantifying flood risk from false data injection attacks	Assumes 100% FDI occurrence; limited to single detector; hypothetical rainfall conditions
12	Kovács (2024)	Model	Multidisciplinary analysis linking water cybersecurity vulnerabilities to human migration	No specific implementation strategies; migration–cybersecurity interrelation insufficiently explored
13	Szádeczky (2023)	Model	Literature review and data analysis of cyber risks in Hungarian/Central European water infrastructure	Weak empirical foundation; no detailed implementation framework provided
14	Murillo et al. (2022)	Model	BATADAL 2.0 dataset (DHALSIM) integrating hydraulic and network traffic data for IDS development	Limited attack type diversity; restricted scenario complexity

foci, whilst the intersection between cybersecurity and sustainability remains underexplored. Notably, no single study within this axis proposes an integrated solution that simultaneously addresses cybersecurity, sustainability, and AI.

Three contributions illustrate the diversity of approaches within this axis. Batarseh et al. (2023) developed ACWA, a testbed that couples artificial intelligence modules with a digital twin of a water distribution network. The testbed simulates operational and adversarial scenarios, enabling researchers to validate detection algorithms against realistic hydraulic dynamics. Its modular design permits the incremental integration of sensor types and attack profiles; nonetheless, the chemical composition of water—a variable that directly affects public health risk—is not captured by the current simulator, restricting the range of attack scenarios that can be meaningfully replicated.

The open-access data policy adopted by the authors partially offsets this limitation by facilitating independent extension of the dataset.

At the governance level, Muttaqin and Kalamullah Ramli (2023) proposed a security framework for the Indonesian water sector that reconciles international standards (C2M2, ISO 27019) with domestic regulation (Government Regulation No. 71 of 2019). The framework organises security controls into four categories—governance and ecosystem, protection, defence, and resilience—and provides a maturity-based roadmap for utilities with limited prior cybersecurity investment. The absence of sector-specific benchmarks against which the framework's effectiveness could be measured represents the principal limitation, since no comparable instrument had previously been constructed for Indonesian water utilities.

Murillo et al. (2022) approached the problem from a modelling perspective, introducing BATADAL 2.0, a dataset constructed with the DHALSIM simulator that integrates hydraulic process data with network traffic traces. By superimposing cyber-attack signatures onto normal operational profiles, the dataset enables the training and benchmarking of intrusion detection systems under controlled conditions. The co-registration of process and network data constitutes the principal methodological advance; the attack repertoire, however, remains narrow relative to the threat landscape documented in recent advisories, and the single-detector assumption constrains the generalisability of detection performance claims.

3.2 Sustainability

The Sustainability axis comprises 14 studies examining how smart technologies contribute to the long-term viability of water networks. Although many reference specific technologies—IoT, SCADA, Water 4.0—the principal focus remains on sustainability outcomes: resource conservation, energy efficiency, and alignment with development goals. Three processes, nine frameworks, and two models were identified.

Table 3 summarises the reviewed studies.

The sustainability studies are predominantly framework-oriented (9 of 14), reflecting the field's current emphasis on structuring guiding principles rather than delivering deployable tools. IoT-based monitoring and smart sensor deployment are the most recurrent technological enablers, whilst energy recovery and water loss reduction constitute the primary sustainability targets. A notable methodological pattern is the reliance on literature reviews and multi-criteria decision analysis (AHP, fuzzy Delphi) rather than longitudinal empirical studies. Geographic bias is apparent: most case studies originate from Europe and Asia, with developing regions—where water stress is most acute—remaining underrepresented. Critically, cybersecurity considerations are virtually absent from the sustainability literature, even when IoT and SCADA systems form the technological backbone of the proposed solutions.

Three studies exemplify the principal research orientations within this axis. Ramos et al. (2019) reported a longitudinal case study of a smart water network deployment in which energy savings of 57 GWh and water savings of 100 Mm^3 were achieved over a twelve-year period, accompanied by a reduction exceeding 47 000 tonnes of CO_2 emissions. The study demonstrated that coupling real-time decision-making models with advanced metering infrastructure can yield quantifiable environmental benefits; the

Table 3 Summary of reviewed studies—Sustainability axis

#	Reference	Type	Key Contribution	Limitations
1	Palermo et al. (2022)	Process	IoT-based real-time monitoring of water levels, consumption, and leak detection in buildings	Limited detail on IoT component integration; insufficient discussion of future challenges
2	Ramos et al. (2019)	Process	Case study achieving 57 GWh energy and 100 Mm^3 water savings over 12 years with 47 000 t CO_2 reduction	Results confined to a single distribution system; limited urban context generalisability
3	Józefowicz and Michniewicz-Ankiersztajn (2023)	Process	Digital tools for rural water management promoting transparency and citizen participation	No detail on technology deployment in varied rural settings; equity and access gaps persist
4	Ramesh et al. (2020)	Framework	IoT and WSN integration for sustainable communities; patented landslide early warning system	Lacks real-time data integration; energy limitations affect sensor durability
5	Liu et al. (2021)	Framework	Fuzzy Delphi identification of 22 barriers to circular economy in smart water management (Zhejiang)	Regional applicability limited; no concrete strategies for overcoming identified barriers
6	Grigg (2022)	Framework	Economic framework for integrated urban water systems promoting sustainability in smart cities	Insufficient depth on financing in decentralised governance; user acceptance not analysed
7	Bouramdane (2023)	Framework	AHP-based multi-criteria decision framework for water management strategy prioritisation	Limited evaluation criteria; subjective weighting; general framework not city-specific
8	Vani et al. (2024)	Framework	IoT sensors for real-time monitoring of pressure, quality, and leaks to promote sustainability	Data privacy and network vulnerability concerns; scalability and interoperability challenges
9	Adedeji et al. (2022)	Framework	Water 4.0 concept applying Industry 4.0 principles (IoT, AI) to digitalise water supply systems	Existing ICT solutions insufficient for system complexity; lacks concrete adoption case studies
10	Okoli and Kabaso (2024)	Framework	PRISMA-based systematic review of IoT in urban water management; identifies 5G research gap	Geographic bias towards Asia; English-language post-2005 filter excludes earlier studies
11	Joseph et al. (2022)	Framework	IWN framework with intelligent algorithms, sensors, and SCADA aligned with SDGs	Economic feasibility of smart technologies in developing economies not validated
12	Giudicianni et al. (2020)	Framework	Energy recovery strategies (microturbines, pumps-as-turbines) with digital twin and blockchain	Limited discussion of practical implementation challenges and infrastructure adaptation costs
13	Mutchek and Williams (2014)	Model	Integrated smart water grid vision applying ICT at multiple levels for monitoring and control	Lack of R&D investment due to institutional inertia; no regulatory framework identified
14	Bonoli et al. (2019)	Model	LCA-based urban metabolism model (GSTMM) evaluating greywater reuse with photovoltaic energy	Omits material analysis for system production; limited to a single urban context

results are, however, drawn from a single distribution system, and their transferability to networks of different scale, topology, or climatic context has not been validated. The case nonetheless provides one of the few empirical baselines against which future smart water deployments can be compared.

Okoli and Kabaso (2024) conducted a PRISMA-based review of IoT applications in urban water management, identifying a persistent research gap around the role of 5G connectivity in enabling low-latency sensor communication for leak detection and pressure regulation. Their bibliometric analysis revealed a geographical concentration of studies in Asia and Western Europe, with sub-Saharan Africa and Latin America substantially underrepresented despite being the regions where water stress is most acute. The review called for longitudinal field trials in resource-constrained settings, a recommendation that directly motivates the phased deployment model proposed within TRITON.

Bonoli et al. (2019) adopted a life-cycle assessment methodology to evaluate a grey-water reuse scheme powered by photovoltaic energy in an urban residential context. Their Green Smart Technology for Water model quantified the environmental trade-offs between centralised treatment and decentralised on-site reuse, concluding that the latter yields lower net energy consumption when solar irradiance exceeds a threshold that depends on building occupancy. The calibration is specific to Mediterranean climatic conditions, and the material analysis for system construction was excluded from the assessment boundary, which limits the completeness of the environmental accounting.

3.3 Artificial Intelligence

The Artificial Intelligence axis encompasses 14 studies that apply AI and machine learning techniques to water network management. Although these works may reference IoT, SCADA, or standards such as ISO 14046, the principal objective is the exploration of AI-based solutions for improving and optimising water resource management. Two processes, seven frameworks, and six models were identified.

Table 4 presents the synthesis.

The AI axis exhibits the most balanced distribution across sub-categories, with models (6) and frameworks (7) roughly equal in number. Leak detection and demand forecasting are the two dominant application areas, addressed through neural networks, pattern recognition, and deep learning. A methodological shift is observable: earlier studies rely on classical machine learning (MLP, Kohonen), whereas more recent contributions employ deep learning (CNN, CVAE) and reinforcement learning. Despite strong performance metrics—several models report detection accuracies exceeding 95%—a persistent limitation is the dependence on simulation-based validation rather than deployment in operational networks. Data quality and model reproducibility remain systemic concerns across the axis. As with the sustainability literature, explicit cybersecurity integration is rare: only one AI study Addeen et al. (2024) directly targets cyber-physical attack detection, underscoring the siloed nature of current research.

Three contributions from this axis merit closer examination. Mounce et al. (2015) developed AURA-Alert, a cloud-based platform for automated anomaly detection in water distribution networks. The system processes time-series data from pressure and flow sensors, applying statistical control charts and machine-learning classifiers to distinguish genuine anomalies from sensor noise. The authors estimated potential cost savings of US$ 12.5 billion per annum if such technology were deployed across the global water sector; practical adoption, nonetheless, hinges on utility willingness to migrate operational data to cloud environments and on the availability of sufficiently granular historical datasets for model training and calibration.

Sharma and Mohan (2023) examined the applicability of Trustworthy AI principles—as defined by the European Commission's High-Level Expert Group—to smart water distribution systems. The study mapped seven requirements (human agency, technical robustness, privacy, transparency, diversity, societal well-being, and accountability) onto typical water network AI use cases, identifying transparency and data governance as the dimensions with the largest implementation deficit. The

Table 4 Summary of AI studies. Studies marked with † contribute to two sub-categories (Process and Framework) and appear twice; the count of 15 rows corresponds to 14 unique publications.

#	Reference	Type	Key Contribution	Limitations
1	Mounce et al. (2015)	Process	Cloud-based AURA-Alert system for anomaly detection in smart water networks; potential $12.5B annual savings	Practical implementation and user acceptance challenges; requires robust data tools
2	Rojek and Studzinski (2019)†	Process	MLP and Kohonen neural networks for leak detection and localisation in Polish water networks	Dependent on input data quality and hydraulic model calibration; no cited standards
3	Tosin Michael Olatunde et al. (2024)	Framework	Review of smart water management in Africa and the US; regulatory analysis (Safe Drinking Water Act)	Financial and technical barriers in Africa; ageing US infrastructure; lack of standardised regulations
4	Lowe et al. (2022)	Framework	Review of AI/ML in water treatment (chlorination, adsorption, membrane filtration, monitoring)	Poor data management and low reproducibility; lack of code transparency
5	Makropoulos and Bouziotas (2023)	Framework	Reinforcement learning agent integrated with UWOT for decentralised water infrastructure planning	RL algorithms do not always converge; water quality simulation insufficient
6	Sharma and Mohan (2023)	Framework	Trustworthy AI (TAI) principles for smart water distribution systems (HLEG guidelines)	No comprehensive TAI framework for varied contexts; infrastructure gaps in developing regions
7	Asian Development Bank et al. (2020)	Framework	AI and big data analytics for unaccounted-for water management in water utilities	Limited discussion of utility-level implementation challenges; organisational resistance unaddressed
8	Rojek and Studzinski (2019)†	Framework	ML-based multi-model control supervisor for metropolitan water distribution monitoring	Data overload challenges; reliance on simulations may reduce real-world applicability
9	Joseph et al. (2022)	Framework	Smart water network framework integrating algorithms, sensors, and SCADA for leak detection	Needs cost-benefit analysis; lack of financial and human resources cited as barrier
10	Addeen et al. (2024)	Model	CVAE-based cyber-physical attack detection in WDS achieving 98% accuracy (NIST-aligned)	Detection accuracy and processing time need improvement; integration with deeper architectures pending
11	Preciado et al. (2019)	Model	High-frequency (1-minute) water demand forecasting using pattern recognition	Untested across diverse geographies; low-density areas may show data inconsistency
12	Benítez et al. (2019)	Model	Pattern similarity-based 24-hour demand forecasting eliminating annual seasonality (ISO 14046)	Struggles with anomalous days (holidays); flow regularity varies by location
13	Jin et al. (2019)	Model	CNN-based image recognition water metering with NB-IoT (3GPP standards)	Microcontroller specificity limits adaptability; accuracy affected by lighting conditions
14	Addas et al. (2023)	Model	DRL robotic prototype: 26% yield increase, 41% water reduction, 33% less chemical use	Limited terrain mobility; maintenance and durability improvements needed
15	Gao et al. (2022)	Model	AI/IoT-based Intelligent Pressure Regulation Vehicle (IPRV) for water supply networks	Lacks context-specific implementation standards; poor responsiveness to extreme fluctuations

absence of a validated compliance framework for the water sector was highlighted as a barrier to regulatory acceptance, particularly in jurisdictions that are beginning to mandate algorithmic auditing for critical infrastructure decisions.

Addeen et al. (2024) proposed a Conditional Variational Autoencoder architecture for detecting cyber-physical attacks in water distribution systems. Trained on the BATADAL benchmark, the model achieved a detection accuracy of 98%, outperforming several baseline classifiers including one-class SVM and isolation forest. The NIST-aligned evaluation protocol lends credibility to the reported metrics; the authors nonetheless noted that processing latency and integration with deeper architectures require investigation before the approach can be considered deployment-ready. This

study stands as one of the few within the AI axis that explicitly targets cybersecurity objectives, underscoring the disciplinary boundaries that characterise the current literature.

4 Results Analysis

This section presents a detailed analysis derived from the systematic review, structured to progress from domain-specific findings to cross-domain synthesis. Each of the three core thematic axes—Cybersecurity, Sustainability, and Artificial Intelligence—is examined through a set of guiding questions designed to elicit the state of the art and the principal gaps within each field. The guiding-question approach follows the methodological recommendations of Kitchenham and Charters for systematic reviews in software engineering, adapted here to the interdisciplinary context of smart water networks. For each axis, a glossary of recurring characteristics is compiled and their presence across the reviewed studies is mapped in a comparative table. The per-axis analyses are then followed by a cross-axis examination that quantifies the extent to which existing proposals address multiple domains simultaneously, thereby revealing the integration deficit that motivates the TRITON framework.

4.1 Cybersecurity Analysis

After reviewing each study, four guiding questions were formulated around the cybersecurity axis. These questions, along with their justifications, are presented below. A summary table consolidating all results is provided at the end.

- Question 1. What cybersecurity technologies have been implemented in smart water networks? Justification: Given the increasing connectivity of water networks, it is important to identify the cybersecurity technologies deployed to protect them. These include solutions such as encryption, intrusion detection, and authentication, which are critical to ensuring data integrity and operational continuity.
- Question 2. What specific vulnerabilities have been identified in these networks? Justification: Smart water networks face particular threats due to their reliance on SCADA systems and IoT sensors. Analysing these vulnerabilities helps to understand the most frequent risks and supports the design of effective mitigation strategies.
- Question 3. Are there applicable standards or regulations for data protection in water networks? Justification: Awareness of standards such as ISO 27001 and local regulations is essential to assess whether cybersecurity solutions align with legal and ethical frameworks, ensuring responsible and secure adoption.
- Question 4. What is the impact of cybersecurity on the sustainability of these networks? Justification: Cybersecurity not only safeguards operations but also indirectly supports sustainability by preventing disruptions and optimising resource management.

After reviewing the literature related to cybersecurity and addressing the above questions, the following findings can be summarised:

Regarding the technologies implemented (Question 1), a wide range of approaches is observed. Several studies highlight the use of Intrusion Detection Systems (IDS), firewalls, and Security Information and Event Management (SIEM) systems. These are crucial for protecting critical water infrastructures, as they allow real-time threat detection and response. The implementation of encryption technologies and multi-factor authentication is also noted, securing both the integrity and confidentiality of sensitive data, particularly in contexts vulnerable to cyberattacks.

With regard to the vulnerabilities identified (Question 2), a common pattern is the lack of software updates Tuptuk et al. (2021) and inadequate cybersecurity training among personnel Szádeczky (2023). These weaknesses are concerning, as they may be exploited to compromise water monitoring and control systems. Furthermore, exposure to denial-of-service (DDoS) attacks and the absence of incident response protocols highlight the urgent need for proactive risk management and a stronger culture of cybersecurity in the sector.

In terms of standards and regulations (Question 3), several studies Ali (2021); Bosco et al. (2022); Muttaqin and Kalamullah Ramli (2023); Szádeczky (2023) refer to well-established frameworks such as ISO 27001, NIST guidelines, and local cybersecurity laws. These serve as essential references for guiding the implementation of effective security measures. Adhering to these frameworks not only helps mitigate cyber risks but also builds public trust in water infrastructure—vital for the long-term sustainability of such systems.

Finally, concerning the impact of cybersecurity on sustainability (Question 4), the use of secure technologies and improved staff training has a positive influence on operational resilience and system efficiency. Better incident response capabilities and continuous service availability contribute to sustainability. The integration of real-time data analytics and IoT solutions is mentioned as a key enabler for more sustainable water resource management.

Taken together, the responses to these four guiding questions reveal a field in which technological solutions are maturing faster than the governance structures required to deploy them. Intrusion detection and encryption technologies exist in prototype form, yet water utilities—particularly smaller ones—lack the organisational capacity and budgetary allocation to adopt them systematically. The gap between technological availability and operational uptake is compounded by a workforce deficit: cybersecurity expertise remains scarce in the water sector, where operational technology staff have traditionally been trained in hydraulic engineering rather than information security. Bridging this gap demands not only technological transfer but also sustained investment in interdisciplinary training programmes, a dimension that the TRITON framework addresses through its Quality Assurance (CI-9) and Incident Management (CI-8) components.

From the literature review, several recurring cybersecurity features have been identified. A glossary of these features is presented below, and Table 5 outlines whether each feature is fully present, partially present, or absent in the reviewed studies. It should be noted that the comparative tables (Tables 5, 6, and 7) are not restricted to

Table 5 Comparative Table: Article vs Presence of Features

Art	AUT	ENC	AUTD	MON	INT	DISP	ESC	APT	ANM	RESP
Szádeczky (2023)	Y	N	Y	Y	N	Y	N	N	P	P
Clark and Panguluri (2018)	P	Y	N	N	Y	Y	Y	P	N	Y
Bosco et al. (2022)	Y	Y	N	Y	Y	N	Y	Y	Y	P
Muttaqin and Kalamullah Ramli (2023)	Y	N	P	Y	Y	Y	N	N	Y	Y
Lin et al. (2024)	Y	Y	Y	P	P	Y	Y	Y	P	Y
Bello et al. (2022)	Y	N	Y	Y	Y	Y	P	Y	Y	N
Alghassab (2021)	Y	Y	N	N	Y	N	P	P	Y	N
Kumar et al. (2019)	P	P	N	Y	N	Y	Y	N	N	P
Małka et al. (2022)	Y	Y	N	P	N	Y	P	P	N	Y
Tuptuk et al. (2021)	P	Y	Y	N	N	N	P	Y	Y	Y
Murillo et al. (2022)	Y	Y	Y	N	Y	N	Y	N	Y	P
Kovács (2024)	N	Y	N	P	Y	P	N	N	P	Y
Józefowicz and Michniewicz-Ankiersztajn (2023)	P	Y	Y	P	N	N	Y	Y	N	Y
Adedeji et al. (2022)	P	N	N	Y	Y	N	N	P	Y	N
T	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Y = Complies; N = Does Not Comply; P = Partially; T = TRITON

the studies listed in the corresponding descriptive tables (Tables 2, 3, and 4). When a study primarily classified under one thematic axis addresses features pertinent to another axis—for example, a sustainability-oriented study that discusses access control mechanisms—it is included in the comparative table of the secondary axis to provide a more comprehensive evaluation of feature coverage across the corpus.

i) AUT. Robust authentication (secure identity validation). ii) ENC. Encryption (protection of data in transit and at rest). iii) AUTD. Dynamic authorisation (context-based permission control). iv) MON. Continuous monitoring (real-time threat detection). v) INT. Data integrity (ensuring data is not altered). vi) DISP. Availability (guaranteeing uninterrupted access to systems). vii) ESC. Scalability (ability to support growing users/devices). viii) APT. Protection against Advanced Persistent Threats. ix) ANM. Malware analysis (detection of malicious software). x) RESP. Incident response (strategies to mitigate and resolve attacks).

Figure 2 displays a bibliographic map showing keyword co-occurrence in the reviewed cybersecurity literature. The clusters highlight thematic groupings such as cybersecurity, Internet of Things, and machine learning, revealing their interrelations across studies.

On the other hand, Fig. 3 illustrates the sectors with the highest number of vulnerabilities identified in the literature review. It can be observed that the IT infrastructure sector is the most susceptible to cybersecurity attacks and failures.

4.2 Sustainability Analysis

Following the examination of the selected studies, four key questions were established to guide the sustainability axis. These questions are presented below alongside their justification, followed by a summary table that consolidates the findings (Table 6).

- Question 1: How is sustainability measured in smart water networks? Justification: Sustainability is a central pillar of modern water networks. Understanding

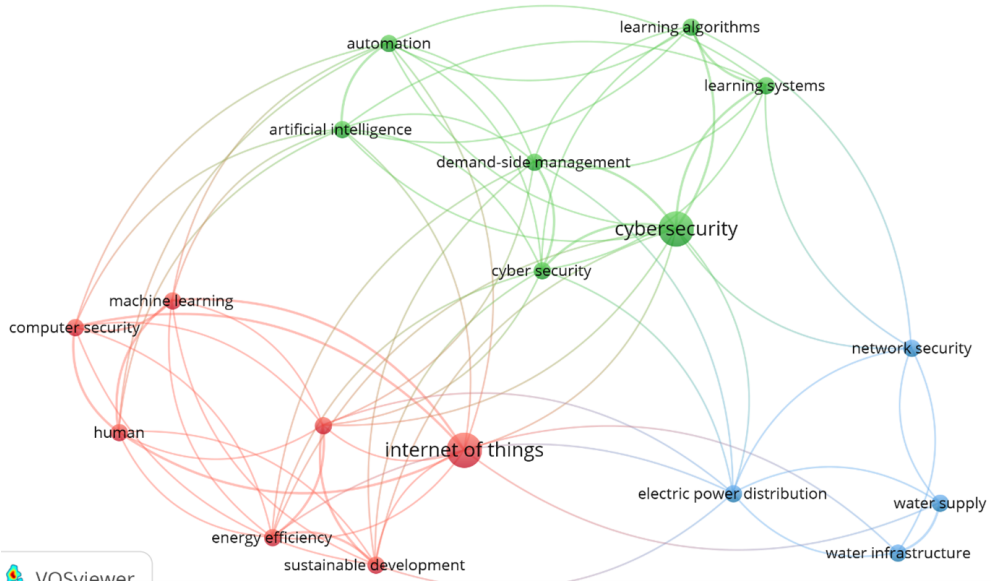


Fig. 2 Bibliographic Map on Cybersecurity

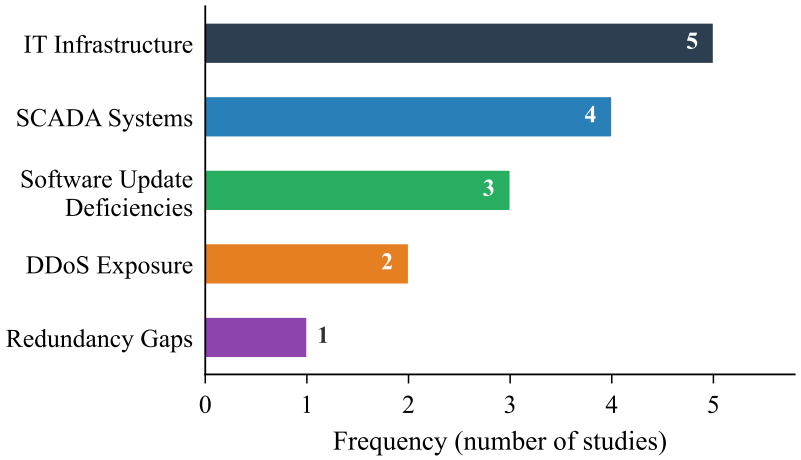


Fig. 3 Most Frequently Reported Vulnerability Categories in the Reviewed Literature

the metrics and evaluation methods is essential to determine whether technological implementations are meeting objectives related to efficiency and resource conservation.

- Question 2: What technologies contribute to energy efficiency and responsible resource use? Justification: Identifying specific technologies—such as smart sensors, data analytics, and monitoring systems—helps clarify how ICTs optimise energy consumption and reduce water waste.

- Question 3: What international standards apply to sustainability in water networks? Justification: Analysing standards such as ISO 14001 or Sustainable Development Goal (SDG) 6 provides a framework for aligning technological practices with globally recognised principles.
- Question 4: How are ICTs integrated to promote sustainability? Justification: ICTs drive smart water networks. Investigating how they are practically integrated helps to identify strategies that effectively enhance sustainability.

After establishing these questions, the findings from the literature review on sustainability in water management were analysed. The results show that a variety of technologies, applications, and international standards contribute to improving resource efficiency. Notably, sustainability is often measured using indicators such as water use efficiency and waste reduction. Innovative technologies such as rainwater harvesting systems, wastewater treatment, and IoT sensors enable more effective water monitoring and management. References are also made to standards like ISO 14001 (environmental management), ISO 50001 (energy management), and UN water guidelines.

The reviewed studies also emphasise the vital role of Information and Communication Technologies (ICTs) in water management. These technologies are integrated via real-time data monitoring and management platforms, which support informed and efficient decision-making. Examples include smart meters and data management systems that help reduce water losses and improve urban water supply services. The automation of processes and data collection is a critical task to optimise water management and ensure responsible resource usage.

Additionally, the application of smart water management systems is discussed, which enables continuous evaluation of sustainability through efficient water distribution and leakage reduction. The implementation of circular economy approaches is also highlighted, aiming to minimise waste and maximise resource use through digitalisation and data analytics.

In conclusion, the integration of sensor networks and digital platforms for efficient water management in smart cities is essential. Sustainability is measured not only by reduced losses and efficient resource use but also by the ability of cities to adapt to current environmental and social challenges. Collectively, the reviewed studies paint a picture of a field in which the technological building blocks for sustainable water management are available but remain unevenly distributed. Advanced metering infrastructure, real-time monitoring platforms, and energy recovery devices have been validated in pilot deployments across Europe and East Asia, yet their reported adoption in low- and middle-income countries remains limited. Cost remains a decisive barrier: the capital expenditure associated with sensor networks and data management platforms exceeds the budgetary capacity of many utilities in developing regions, and the business case for smart water investment is often undermined by tariff structures that do not reflect the true cost of service provision. The circular economy paradigm offers a conceptual bridge between financial sustainability and environmental sustainability, but its practical implementation in the water sector requires regulatory reform and institutional capacity that few developing-country utilities currently possess. The

Table 6 Comparative Table: Sustainability Characteristics

Art	EUR	MTR	RP	ER	GIA	R	AC	IAR	EC
Mutchek and Williams (2014)	N	Y	N	P	N	Y	N	N	P
Giudicianni et al. (2020)	N	N	Y	N	N	P	Y	N	N
Ramesh et al. (2020)	Y	P	N	N	N	N	N	P	Y
Ramos et al. (2019)	N	Y	Y	P	Y	N	N	N	Y
Liu et al. (2021)	N	Y	N	P	P	N	N	N	P
Bouramdane (2023)	Y	N	P	Y	N	N	Y	N	N
Joseph et al. (2022)	N	Y	N	P	N	N	P	N	Y
Visser et al. (2021)	N	N	Y	P	Y	N	N	N	P
Grigg (2022)	N	Y	N	N	P	Y	N	P	N
Okoli and Kabaso (2024)	P	N	N	Y	N	P	P	N	Y
Bonoli et al. (2019)	Y	Y	N	P	P	N	N	P	N
Fengchun (2021)	P	N	N	N	N	Y	P	N	Y
Addeen et al. (2024)	Y	N	P	N	Y	N	N	P	P
Lowe et al. (2022)	P	Y	N	N	N	Y	N	Y	N
T	Y	Y	Y	Y	Y	Y	Y	Y	Y

Y = Complies; N = Does Not Comply; P = Partially; T = TRITON

TRITON framework accounts for this asymmetry through a phased deployment model (Section 5) that sequences investments according to the utility's maturity level, prioritising loss reduction and basic monitoring before more advanced sustainability metrics are pursued.

A set of recurring sustainability characteristics was identified across the literature. The glossary below outlines these characteristics, while their presence or partial implementation across articles is mapped in Table 6.

i) EUR – Efficient Resource Use (Optimisation of water and energy consumption). ii) MTR – Real-Time Monitoring (Sensor deployment for real-time data acquisition). iii) RP – Loss Reduction (Minimisation of leaks or water waste). iv) ER – Renewable Energy (Use of sustainable energy sources in operations). v) GIA – Intelligent Demand Management (Data-driven forecasting and demand management). vi) R – Resilience (System adaptability to changes or disruptions). vii) AC – Automation and Control (Integration of algorithms and automatic systems for process optimisation). viii) IAR – Reduced Environmental Impact (Minimisation of ecological footprint). ix) EC – Education and Awareness (Strategies to engage users in sustainability efforts).

In Fig. 4, the knowledge map reveals three interconnected research themes. The primary cluster centres on the core concepts of sustainability, water resource, and water management, linking them to direct impacts like water footprint and ecosystems. A second theme expands on sustainable development, connecting it to applied challenges such as climate change and watershed management, while a third highlights the socio-political dimension by linking water supply and demand with stakeholders. Overall, the map demonstrates an integrated field where technical water management is inseparable from the broader frameworks of sustainable development and governance.

In turn, Figure 5, shows the frequency of different technologies for energy efficiency and responsible use, as identified in the literature review. The Internet of Things (IoT) stands out as the predominant technology, followed by monitoring sensors, which

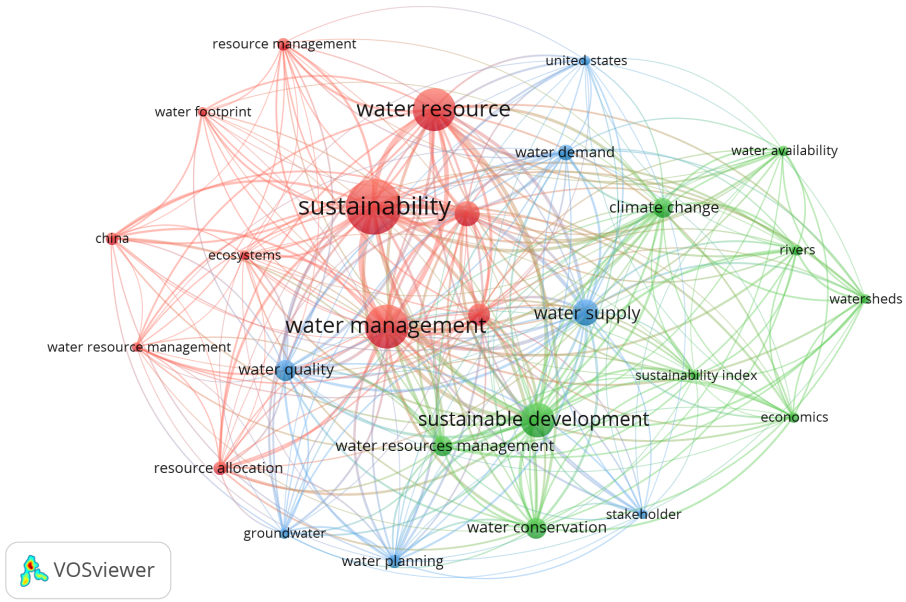


Fig. 4 Bibliographic Map of Sustainability

indicates a clear trend towards digitalisation and real-time data capture to optimise resource management.

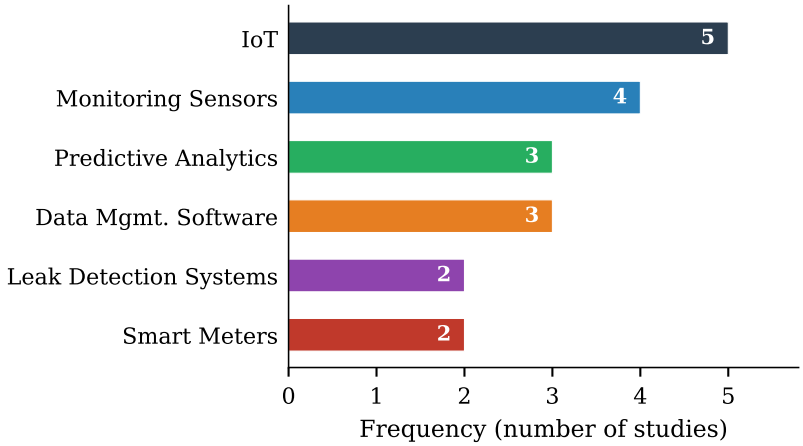


Fig. 5 Technologies Applied in Energy Efficiency

4.3 Artificial Intelligence Analysis

Following the analysis of each reviewed study, four key questions were formulated around the axis of artificial intelligence. These are presented below, along with their justification, and a summary table consolidating the results.

- Question 1. What AI models have been used in the management of smart water networks? Justification: AI is key to enhancing prediction, analysis, and decision-making in water networks. Identifying the models employed, such as machine learning or neural networks, helps to understand their impact on water resource management.
- Question 2. What specific problems does AI solve in this context? Justification: Analysing the issues addressed by AI, such as leak detection or consumption optimisation, helps to assess its relevance and effectiveness in the sector.
- Question 3. Are there legal or ethical frameworks regulating the use of AI in water networks? Justification: The regulation of AI is an emerging topic. Identifying legal and ethical frameworks ensures that its implementation is responsible and does not compromise users' privacy or rights.
- Question 4. How do AI solutions contribute to the efficiency and sustainability of water networks? Justification: AI can be a critical enabler in achieving sustainability and efficiency objectives. Understanding this contribution supports its adoption in smart water projects.

Once the questions were established, the following is an analysis of the results obtained from the reviewed studies.

After a detailed analysis, it is generally highlighted that Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs) are frequently mentioned. These are used for digit recognition in water meters and classification of water usage, as well as for flow prediction in multiphase meters. These technologies enable automated readings, reducing manual workload and improving reading accuracy, thereby contributing to more efficient water management.

Furthermore, reinforcement learning models, such as the Proximal Policy Optimization (PPO) algorithm, are explored for optimising water resource management, achieving a notable reduction of between 20% and 30% in water usage in some cases. Other technologies such as SCADA and big data analytics are also highlighted, allowing real-time monitoring and control of water networks, facilitating more efficient pressure management without compromising customer consumption. Another approach mentioned is the use of anomaly detection algorithms, such as the Conditional Variational Autoencoder (CVAE), which are employed to identify cyberattacks and data manipulation in water distribution systems.

The reviewed works also address the importance of optimisation in the planning and management of decentralised water infrastructures, using intelligent planning agents based on reinforcement learning. This approach is crucial for solving challenges in off-grid environments, where water management can be particularly demanding. Additionally, optimisation techniques are referenced that address issues such as water scarcity, efficiency in distribution, and improvement in water quality.

However, despite the advances in implementing AI for water management, some studies point out that no specific legal or ethical frameworks are mentioned to regulate its use. The need to establish regulations that support the implementation of intelligent technologies and safeguard water resources is emphasised. Overall, the studies present an integrated vision of how artificial intelligence is transforming the management of water networks, highlighting both its benefits and the need for an appropriate regulatory framework to ensure responsible and sustainable use of these technologies.

The responses to these guiding questions expose a tension between algorithmic sophistication and operational maturity. Deep learning and reinforcement learning architectures achieve high accuracy on benchmark datasets, yet the water sector's data infrastructure is seldom adequate to support such models in production. Sensor drift, intermittent telemetry, and legacy SCADA protocols introduce noise and latency that degrade model performance in ways not captured by laboratory evaluations. Data governance presents a further challenge: consumption records and network topology data are frequently held by different departments or agencies, and the absence of standardised data-sharing agreements impedes the construction of training datasets of sufficient volume and diversity. Federated learning, in which models are trained across distributed datasets without centralising raw records, offers a promising direction for overcoming data silos whilst preserving privacy—a capability that the TRITON framework accommodates through the joint operation of its AI and Cybersecurity layers. The regulatory vacuum identified across the reviewed studies amplifies these concerns: without sector-specific AI governance standards, utilities lack a compliance baseline against which to evaluate algorithmic transparency, fairness, and accountability.

On the other hand, after reviewing the literature, several characteristics related to artificial intelligence were identified. Below is a glossary describing each characteristic, while Table 7 maps their presence or absence in each of the reviewed articles.

i) DA: Anomaly Detection (Automatic identification of leaks, consumption irregularities, or system failures). ii) PM: Predictive Maintenance (Use of predictive models to anticipate infrastructure failures and plan repairs). iii) OP: Process Optimisation (Application of algorithms to enhance operational efficiency, such as water distribution or energy consumption reduction). iv) AA: Advanced Automation (Implementation of AI to carry out complex tasks without constant human intervention). v) AP: Predictive Analytics (Use of historical and real-time data to forecast future behaviour in water networks). vi) TA: Autonomous Decision-Making (AI systems making real-time decisions based on data and predefined rules). vii) PI: Intelligent Personalisation (Adapting water supply and management to the specific needs of different users or regions). viii) IAO: Integration with Other Operations (Interconnection with other intelligent systems, such as electric or transport networks, to enhance overall efficiency). ix) UML: Use of Legal Frameworks (Application of laws or standards regulating the use of AI in smart water networks).

In Fig. 6, the knowledge map based on keyword co-occurrence in the reviewed literature. It highlights thematic clusters related to AI in water networks, such as water management, smart water systems, and optimisation methods. Node size reflects keyword frequency, while colour indicates topic grouping.

Table 7 Comparative Table: Characteristics in Artificial Intelligence

Ref	DA	PM	OP	AA	AP	TA	PI	IAO	UML
Tosin Michael Olatunde et al. (2024)	N	Y	N	Y	N	N	Y	P	N
Makropoulos and Bouziotas (2023)	Y	N	Y	P	N	N	N	Y	N
Sharma and Mohan (2023)	N	Y	P	N	Y	N	Y	N	P
Asian Development Bank et al. (2020)	N	N	Y	N	N	Y	N	N	N
Mounce et al. (2015)	Y	P	N	N	P	N	N	Y	N
Rojek and Studzinski (2019)	N	Y	N	P	Y	N	N	N	N
Preciado et al. (2019)	P	N	Y	N	N	N	N	P	N
Predescu et al. (2020)	N	Y	N	N	N	P	P	Y	N
Benítez et al. (2019)	Y	N	P	Y	N	N	N	P	N
Jin et al. (2019)	N	P	N	N	P	P	Y	N	N
Joseph et al. (2022)	N	N	Y	N	N	N	P	Y	Y
Addas et al. (2023)	Y	N	P	N	N	N	N	N	P
Gao et al. (2022)	N	Y	N	Y	P	Y	N	N	P
T	Y	Y	Y	Y	Y	Y	Y	Y	Y

Y = Complies; N = Does Not Comply; P = Partially; T = TRITON

In Fig. 7, the frequency with which legal frameworks are addressed in studies on the use of AI in water networks is shown. It reveals that most publications do not specify a regulatory basis, while only a few refer to regional policies or highlight the need for clearer regulations. This indicates a significant regulatory gap in the responsible implementation of AI technologies in the water sector.

4.4 Integrated Analysis

The analysis conducted on the axes of cybersecurity, sustainability, and artificial intelligence in smart water networks highlights their relevance and the interdependence that exists among these domains. Cybersecurity aims to ensure the integrity, confidentiality, and availability of the data generated by intelligent technologies, protecting networks from potential threats. Sustainability, in turn, focuses on optimising the use of water resources, reducing losses, and promoting best practices in water management. Artificial intelligence acts as an enabler, allowing the optimisation of processes such as demand prediction, leak detection, and real-time management, thereby enhancing both the efficiency and sustainability of these networks.

However, although each of these axes contributes significantly, not all reviewed articles simultaneously integrate these three approaches into a single framework.

In the analysed articles, it is observed that most address one or two of the main axes, such as the implementation of AI in sustainable solutions or the strengthening of cybersecurity in smart networks. Nonetheless, none of the reviewed studies manages to comprehensively address all three pillars. Among the two-axis combinations, the pairing of sustainability and artificial intelligence is the most frequent, appearing in studies that employ AI for demand forecasting, leak detection, or energy optimisation—all with explicit sustainability objectives. The cybersecurity–AI pairing is less common and is typically limited to intrusion detection in SCADA systems, where the AI component serves a defensive function within a cybersecurity context. The cybersecurity–sustainability pairing is the rarest two-axis combination, found

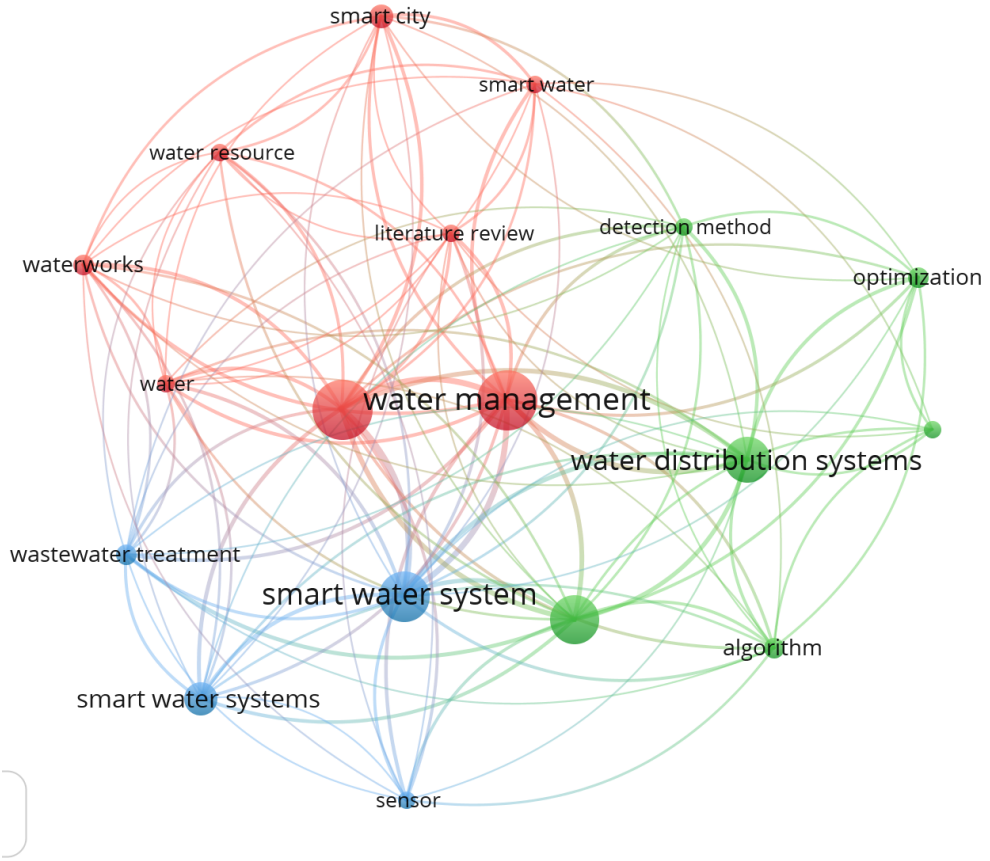


Fig. 6 Bibliographic Map of AI

only in studies that discuss the protection of IoT-enabled monitoring platforms without proposing AI-based analytics. The quantitative distribution of axis coverage is examined below.

Quantitatively, the cross-axis mapping reveals the extent of this disciplinary isolation. Of the 40 reviewed studies, 22 (55%) address a single thematic axis, 18 (45%) incorporate elements from two axes, and none of the studies included in the final corpus was coded as providing substantive coverage of all three simultaneously. This distribution pattern—where the majority of contributions remain confined to one domain—is consistent with the fragmentation documented in recent systematic reviews on WDN cybersecurity (Habib et al. 2026) and anomaly detection in water distribution systems (Galarza Yallico and Santos López 2024). Parallel developments in digital twin architectures for water infrastructure (Ghorbani Bam et al. 2025) and smart water system design (Quintana et al. 2025) indicate a growing recognition that operational,

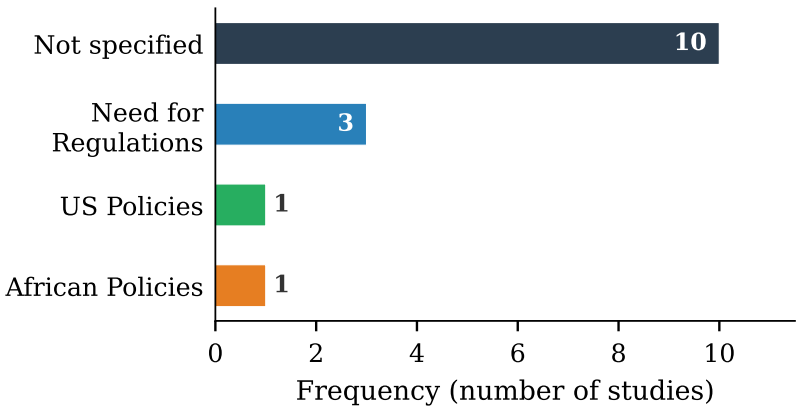


Fig. 7 Legal Frameworks for the Use of Artificial Intelligence

environmental, and security concerns must be treated jointly; however, neither body of work achieves the three-pillar convergence that TRITON proposes. From the governance perspective, the NIST Cybersecurity Framework 2.0 (Pascoe et al. 2024) and the CISA Zero Trust guidance for operational technology environments (Cybersecurity and Infrastructure Security Agency (CISA) 2026) represent substantive advances in securing critical infrastructure, yet both instruments remain silent on sustainability metrics and AI-driven optimisation. The absence of cross-domain governance mechanisms reinforces the argument that existing standards, though valuable within their scope, do not provide a unified reference for water utilities seeking to balance resilience, resource efficiency, and intelligent automation. TRITON is positioned to address this gap by structuring its architecture around all three axes as co-equal design requirements rather than treating any one of them as ancillary.

Table 8 below presents the summary comparative chart that groups the three main axes (Cybersecurity, Sustainability, and Artificial Intelligence) along with their respective characteristics, contrasted across the reviewed studies.

The fragmented coverage observed across the three axes raises a fundamental question for the field: to what extent can cybersecurity, sustainability, and AI be treated as independent concerns in the design of smart water networks? The evidence assembled in this review suggests that treating them in isolation leads to incomplete solutions. A water network protected by robust cybersecurity controls but lacking energy optimisation is operationally wasteful; conversely, a sustainability-oriented deployment without adequate security safeguards exposes critical infrastructure to threats that could negate the environmental gains. AI, as a transversal enabler, has the potential to link these domains through shared data pipelines and predictive analytics, yet its deployment without cybersecurity guardrails introduces new attack surfaces. The integration deficit identified in this review—no article addresses all three axes—thus represents not merely a bibliometric observation but a substantive gap with practical implications for infrastructure design and policy formulation.

Table 8 Summary of characteristics across thematic axes

Art	Cybersecurity										Sustainability								Artificial Intelligence							
x	AUT	ENC	AUTD	MON	INT	DISP	ESC	APT	ANM	RESP	EUR	MTR	RP	ER	GIA	R	AC	IAR	EC	DA	PM	OP	AA	AP	TA	PI	IAO	UML
Szádeczky (2023)	Y	N	Y	Y	N	Y	N	N	P	P																		
Clark and Panguluri (2018)	P	Y	N	N	Y	Y	Y	P	N	Y																		
Bosco et al. (2022)	Y	Y	N	Y	Y	N	Y	Y	Y	P			Y															
Muttaqin and Kalamullah Ramli (2023)	Y	N	P	Y	Y	Y	N	N	Y	Y												Y						
Lin et al. (2024)	Y	Y	Y	P	P	Y	Y	Y	P	Y																		
Bello et al. (2022)	Y	N	Y	Y	Y	Y	P	Y	Y	N																		
Alghassab (2021)	Y	Y	N	N	Y	N	P	P	Y	N																		
Kumar et al. (2019)	P	P	N	Y	N	Y	Y	N	N	P					Y													
Małka et al. (2022)	Y	Y	N	P	N	Y	P	P	N	Y																		
Tuptuk et al. (2021)	P	Y	Y	N	N	N	P	Y	Y	Y																		
Murillo et al. (2022)	Y	Y	Y	N	Y	N	Y	N	Y	P															Y			
Kovács (2024)	N	Y	N	P	Y	P	N	N	P	Y																		
Józefowicz and Michniewicz-Ankiersztajn (2023)	P	Y	Y	P	N	N	Y	Y	N	Y																		
Adedeji et al. (2022)	P	N	N	Y	Y	N	N	P	Y	N								Y										
Mutchek and Williams (2014)	N	Y	N	P	N	Y	N	N	P	N																		
Giudicianni et al. (2020)											N	N	Y	N	N	P	Y	N	N									
Ramesh et al. (2020)											Y	P	N	N	N	N	N	P	Y						Y			
Ramos et al. (2019)	Y										N	Y	Y	P	Y	N	N	N	Y									
Liu et al. (2021)											N	Y	N	P	P	N	N	N	P									
Bouramdane (2023)											Y	N	P	Y	N	N	Y	N	N									
Joseph et al. (2022)				Y							N	Y	N	P	N	N	P	N	Y									
Visser et al. (2021)											N	N	Y	P	Y	N	N	N	P									
Grigg (2022)									Y		N	Y	N	N	P	Y	N	P	N							Y		
Okoli and Kabaso (2024)											P	N	N	Y	N	P	P	N	Y									
Bonoli et al. (2019)				Y							Y	Y	N	P	P	N	N	P	N		Y							
Fengchun (2021)											P	N	N	N	N	Y	P	N	Y									
Addeen et al. (2024)											Y	N	P	N	Y	N	N	P	P									
Lowe et al. (2022)		Y									P	Y	N	N	N	Y	N	Y	N					Y				
Tosin Michael Olatunde et al. (2024)																				N	Y	N	Y	N	N	Y	P	N
Makropoulos and Bouziotas (2023)																				Y	N	Y	P	N	N	N	Y	N
Sharma and Mohan (2023)																				N	Y	P	N	Y	N	Y	N	P
Asian Development Bank et al. (2020)						Y														N	N	Y	N	N	Y	N	N	N
Mounce et al. (2015)													Y							Y	P	N	N	P	N	N	Y	N
Rojek and Studzinski (2019)																				N	Y	N	P	Y	N	N	N	N
Preciado et al. (2019)																				P	N	Y	N	N	N	N	P	N
Predescu et al. (2020)																				N	Y	N	N	N	P	P	Y	N
Benítez et al. (2019)		Y																		Y	N	P	Y	N	N	N	P	N
Jin et al. (2019)												Y								N	P	N	N	P	P	Y	N	N
Joseph et al. (2022)																	Y			N	N	Y	N	N	N	P	Y	Y
Addas et al. (2023)						Y														Y	N	P	N	N	N	N	N	P
Gao et al. (2022)																				N	Y	N	Y	P	Y	N	N	P
T	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Y = Yes; N = No; P = Partial; T = TRITON

Table 9 Thematic Axis Coverage across Reviewed Articles. Axis-level counts are non-mutually exclusive: one study (Joseph et al.) contributes to both SO and IA. Percentages of single-, two-, and three-axis coverage are calculated independently within each group; the aggregate figures over the 40 unique publications are 22 (55%) single-axis, 18 (45%) two-axis, and 0 three-axis.

Group	No. Papers	1 Axis	2 Axes	3 Axes
SO	14	50%	50%	0%
CI	13	54%	46%	0%
IA	14	57%	43%	0%

A summary analysis of the 40 reviewed studies is presented in Table 9, grouped according to the three main axes: Artificial Intelligence, Cybersecurity, and Sustainability. The table shows the percentage of articles that focus exclusively on one axis, those that integrate two of the axes, and those that simultaneously address all three axes.

Subsequently, Fig. 8 presents, in percentage terms, the occurrence of the thematic axes across the 40 reviewed studies. The figure shows that the majority of studies focus on only one thematic axis, with AI having the highest individual representation.

This demonstrates a fragmented approach in current research, where comprehensive integration across the three axes remains limited.

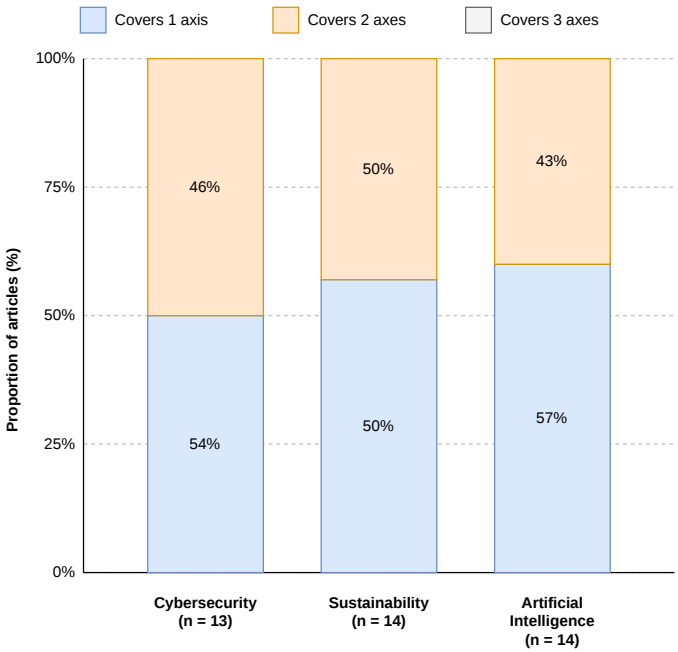


Fig. 8 Presence of Themes within the Thematic Axes

5 TRITON Framework Proposal

The shortcomings identified in the management and monitoring of water networks have served as the foundation for the development of the TRITON framework (Technology Resilience, cybersecurity and Intelligence for Transformative Optimization of smart water Networks). This framework proposes a structured, multi-layered architecture for the application of ICTs, organised around three interdependent pillars: Cybersecurity (CI), Sustainability (SO), and Artificial Intelligence (AI). Tables 10, 11, and 12 describe these thematic axes, outlining the key characteristics that define each pillar. The content of each table is explained below to provide a clearer understanding of the specific contributions of each domain to the TRITON framework. The detailed technical specification of each layer—including algorithms, protocols, normative alignment, and a field-level application—is presented in a companion paper.

Table 10 summarises the cybersecurity dimension, which includes technical measures such as encryption, anomaly detection, and network security, along with strategic

Table 10 TRITON Characteristics in Cybersecurity

Feature	Description
Data Protection (CI-1)	Refers to the implementation of measures to ensure the confidentiality, integrity, and availability of data.
Access Control (CI-2)	Implementation of robust authentication and authorisation systems to limit access to critical systems.
Attack Protection (CI-3)	Use of firewalls, intrusion detection systems (IDS), and other tools to prevent and mitigate cyberattacks.
IoT Device Security (CI-4)	Strengthening security in IoT devices through encryption, authentication, and regular updates.
Encryption (CI-5)	Protection of data in transit and at rest through encryption techniques.
Anomaly Detection (CI-6)	Implementation of systems that monitor system behaviour to detect unusual or malicious activities.
Network Security (CI-7)	Protection of the network infrastructure through firewalls, network segmentation, and other measures.
Incident Management (CI-8)	Development of protocols and response plans to efficiently mitigate and resolve cyber incidents.
Quality Assurance (CI-9)	Implementation of standards and practices to ensure the quality and security of systems.
Real-time Monitoring (CI-10)	Use of tools that enable continuous system monitoring to detect and respond to threats in real time.

Table 11 TRITON Characteristics in Sustainability

Feature	Description
Efficient Water Use (SO-1)	Implementation of technologies and practices that optimise water consumption.
Quality Monitoring (SO-2)	Use of sensors and monitoring systems to ensure water quality.
Water Harvesting and Reuse (SO-3)	Development of systems that enable water harvesting and reuse.
Leak Detection (SO-4)	Implementation of technologies to detect and repair water leaks.
Renewable Energy (SO-5)	Use of sustainable energy sources in water management.
Waste Reduction (SO-6)	Implementation of practices and technologies that minimise water waste.
Energy Consumption Optimisation (SO-7)	Efficient use of energy in water management systems.
Environmental Awareness (SO-8)	Educational and awareness programmes to promote sustainable practices.

mechanisms like access control, incident management, and quality assurance. These elements aim to ensure the integrity, confidentiality, and resilience of smart water infrastructures.

Table 11 presents the sustainability-related components, highlighting features such as efficient water use, integration of renewable energy, leak detection, waste reduction, and environmental awareness. These components are designed to minimise environmental impact while promoting responsible and long-term resource management.

Table 12 outlines the artificial intelligence functionalities incorporated into TRITON. These include consumption forecasting, process automation, real-time analytics, smart alert systems, and pattern recognition. Together, these features enhance operational efficiency, enable data-driven decision-making, and improve system adaptability in dynamic water management environments.

In the literature review, various challenges have been identified for each thematic axis, and Table 13 presents the challenges related to Cybersecurity (CI), associating each one with a possible solution and the corresponding feature.

Table 12 TRITON Characteristics in Artificial Intelligence

Feature	Description
Consumption Prediction (IA-1)	Implement AI models that use historical and contextual data to improve demand forecasting accuracy.
Process Automation (IA-2)	Develop AI-based automated systems for remote meter reading using IoT technologies.
Machine Learning (IA-3)	Improve machine learning algorithms to enhance leak detection accuracy and reduce false positives.
Data-driven Decision-making (IA-4)	Develop AI systems that enable real-time decision-making based on historical and current data.
Real-time Analysis (IA-5)	Implement real-time analysis systems for immediate detection of water quality issues.
Distribution Optimisation (IA-6)	Use AI-based optimisation algorithms to improve water distribution efficiency.
Smart Alerts (IA-7)	Implement smart alert systems that proactively notify about leaks or network failures.
Pattern Recognition (IA-8)	Develop pattern recognition algorithms adaptable to different contexts and population densities.

Similarly, when analysing the studies related to the sustainability axis, several weaknesses were identified, which are presented in Table 14, associating each one with its corresponding feature and potential solution.

Finally, for the Artificial Intelligence axis, a similar procedure was followed by identifying the problems and associating them with potential solutions and the related feature. These relationships are described in Table 15.

In Fig. 9, the architectural structure underlying the proposed TRITON framework is presented in a graphical format.

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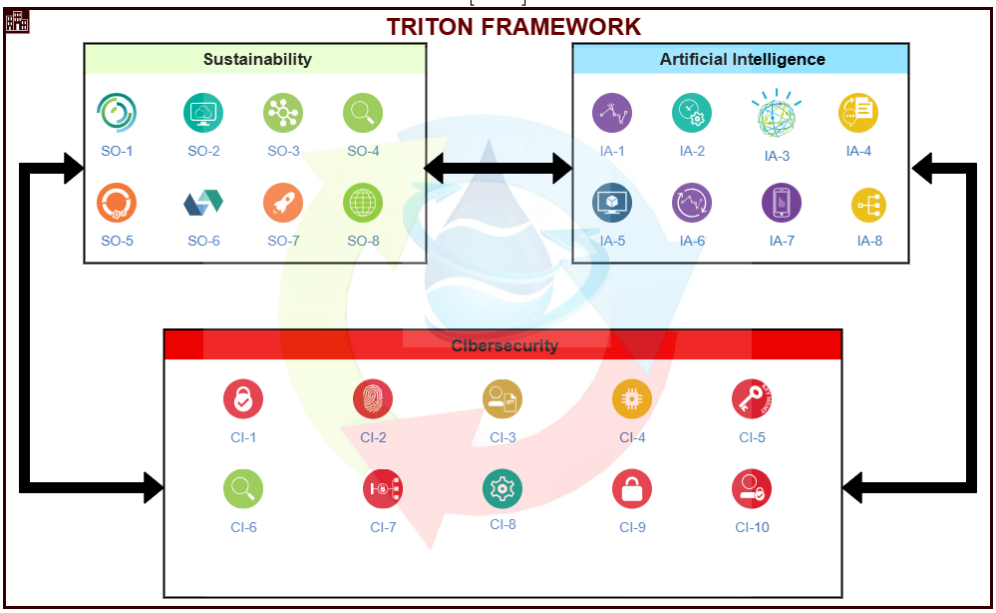


Fig. 9 TRITON Architecture

Table 13 Relationship between challenges and solutions related to cybersecurity

Challenge	Solution	Feature	Evidence
Lack of software updates and insufficient staff training in cybersecurity.	Implement continuous training programmes and automatic software updates.	CI-8 and CI-9	Tuptuk et al. (2021), Szádeczky (2023)
Exposure to denial-of-service (DDoS) attacks and absence of response protocols.	Implement anomaly detection systems and incident response protocols.	CI-6 and CI-8	Bello et al. (2022)
Vulnerabilities in SCADA systems and IoT sensors.	Strengthen security in IoT devices through robust authentication and data encryption.	CI-4, CI-5 and CI-2	Bosco et al. (2022), Muttaqin and Kalamullah Ramli (2023)
Challenges in implementing emerging technologies such as AI and blockchain.	Develop guidelines and standards for the secure implementation of emerging technologies in the water sector.	CI-1, CI-5 and CI-9	Ali (2021)
Limitations in detecting attacks in water distribution systems.	Implement advanced intrusion detection systems and malware analysis.	CI-1, CI-3 and CI-10	Galarza Yallico and Santos López (2024), Habib et al. (2026)

The architecture of the framework is described below, indicating a layered description of the TRITON Framework.

5.1 Cybersecurity Layer

Cybersecurity is a fundamental component of the TRITON framework. As detailed in Table 10, this layer is responsible for protecting both the digital and physical infrastructure against threats and vulnerabilities. Data protection mechanisms and access controls are implemented to ensure that only authorised users can interact with the system. Encryption techniques are also employed to guarantee the confidentiality and security of transmitted information. This layer includes measures to safeguard against cyberattacks, reinforcing IoT devices and communication networks. Additionally, anomaly detection tools and real-time monitoring are incorporated, enabling proactive identification and mitigation of potential threats. To ensure service continuity, incident management and quality assurance protocols are included, allowing for a rapid response to unexpected events. Through this layer, the framework aims to ensure the security and resilience of interconnected systems within smart environments.

Table 14 Relationship between challenges and solutions related to sustainability

Challenge	Solution	Feature	Evidence
Lack of integration of IoT systems in water management, limiting efficiency.	Implement standardised IoT systems to optimise water consumption and management.	SO-01	Gupta et al. (2020)
Limitations in real-time monitoring of water quality.	Deploy advanced sensors and continuous monitoring systems to ensure water quality.	SO-02	Feng et al. (2024)
Limited implementation of water reuse technologies in urban areas.	Promote collection and treatment systems for wastewater reuse in irrigation and other uses.	SO-03	Bonoli et al. (2019)
High levels of undetected leaks in water distribution networks.	Implement early leak detection technologies, such as acoustic sensors and advanced monitoring systems.	SO-04	Giudicianni et al. (2020)
Dependence on non-renewable energy sources in water management systems.	Integrate renewable energy sources (solar, wind) into pumping and water treatment systems.	SO-05	Bouramdane (2023)
Significant water waste due to inefficient practices.	Implement intelligent water management systems to minimise waste and optimise use.	SO-06	Ramos et al. (2019)
High energy consumption in water pumping and treatment systems.	Use energy-efficient technologies and optimised pumping systems to reduce consumption.	SO-07	Quintana et al. (2025)
Lack of environmental awareness in communities regarding sustainable water use.	Develop educational programmes and awareness campaigns to promote sustainable water use practices.	SO-08	Ramesh et al. (2020)

Within operational technology (OT) environments, Zero Trust Architecture has gained traction as a paradigm that eliminates implicit trust assumptions across network boundaries Cybersecurity and Infrastructure Security Agency (CISA) (2026), a direction consistent with the governance functions defined in NIST CSF 2.0 Pascoe et al. (2024). Under this model, CI-2 (Access Control) and CI-7 (Network Security) can be operationalised through micro-segmentation of industrial subnets and continuous identity verification at each access point. Micro-segmentation confines lateral movement in the event of a breach, whilst continuous verification ensures that credentials are re-evaluated against contextual risk indicators rather than granted once at the perimeter. These mechanisms complement the anomaly detection (CI-6) and

Table 15 Relationship between issues and solutions related to AI

Problem	Solution	Feature	Evidence
Difficulty in predicting water demand across different geographical contexts.	Implement AI models that use historical and contextual data to enhance accuracy in demand forecasting.	IA-01	Benítez et al. (2019)
Lack of automation in water meter reading in rural areas.	Develop AI-based automated systems for remote meter reading using technologies such as IoT.	IA-02	Tosin Michael Olatunde et al. (2024)
Limitations in the accuracy of leak detection models.	Improve machine learning algorithms to enhance leak detection accuracy and reduce false positives.	IA-03	Rojek and Studzinski (2019)
Lack of rapid and effective decision-making in emergency situations.	Develop AI systems capable of real-time decision-making based on historical and current data.	IA-04	Makropoulos and Bouziotas (2023)
Limitations in real-time water quality monitoring.	Implement real-time analytics systems to enable immediate detection of water quality issues.	IA-05	Feng et al. (2024)
Inefficiencies in water distribution due to lack of optimisation.	Use AI-based optimisation algorithms to improve efficiency in water distribution.	IA-06	Asian Development Bank et al. (2020)
Lack of early warnings for leaks or failures in the distribution network.	Implement intelligent alert systems that proactively notify of leaks or network failures.	IA-07	Mounce et al. (2015)
Difficulty identifying consumption patterns in low-density population areas.	Develop pattern recognition algorithms that can adapt to different contexts and population densities.	IA-08	Jin et al. (2019)

real-time monitoring (CI-10) capabilities already specified in the layer, forming a defence-in-depth posture suited to distributed water infrastructure.

In operational terms, the cybersecurity layer prescribes a minimum set of controls that must be active before any data-sharing or AI-training activities can commence. Sensor data transmitted from remote sites to a central analytics platform traverses multiple network segments, each of which represents a potential interception point. The layer therefore mandates end-to-end encryption (CI-5) for all telemetry channels and certificate-based device authentication (CI-4) to prevent rogue sensor injection—a threat vector demonstrated in the BATADAL dataset reviewed in Section 3.1 and experimentally validated using the SWaT and WADI testbeds (Goh et al. 2017;

Ahmed et al. 2017). Incident management protocols (CI-8) specify escalation pathways and recovery procedures calibrated to the criticality of the affected subsystem, distinguishing between disruptions to monitoring-only functions and disruptions to actuator-controlled processes such as pump operation or valve adjustment. These controls are aligned with NIST SP 800-82 Rev. 3 (Stouffer et al. 2023), which provides a comprehensive reference for OT security encompassing performance, reliability, and safety constraints specific to industrial environments. From a regulatory standpoint, the European NIS2 Directive (European Parliament and Council of the European Union 2022) now classifies drinking water supply and wastewater management as essential services, mandating risk management, incident reporting, and supply-chain security obligations that CI-8 and CI-9 are designed to support.

A limitation of the current framework specification is the absence of an explicit threat model. Whilst the controls CI-1 through CI-10 address broad categories of risk, a deployment-ready instantiation of TRITON would require a structured adversary characterisation—specifying attacker capabilities (remote vs. physical access, insider vs. external), targeted assets (sensors, PLCs, RTUs, edge gateways, cloud platforms), trust boundaries between IT and OT zones, and the potential consequences of compromise on water quality, supply continuity, and public safety. The MITRE ATT&CK for ICS knowledge base (MITRE Corporation 2024) provides a suitable taxonomy for mapping CI controls against tactics such as Initial Access, Lateral Movement, Inhibit Response Function, and Impair Process Control. Experimental work on cascading effects in interconnected water infrastructure (Palleti et al. 2021) and Bayesian models for distinguishing intentional attacks from accidental failures (Chockalingam et al. 2021) further demonstrate that threat modelling in water systems must account for the interplay between cyber events, physical faults, and human error. The detailed threat model and its mapping to IEC 62443 zones and conduits are deferred to the companion paper.

5.2 Sustainability Layer

As outlined in Table 11, the sustainability layer promotes the efficient use of water and energy, minimising waste and promoting responsible practices. The integration of water quality monitoring and leak detection helps to reduce losses and improve the management of water resources. Additionally, the framework encourages the capture and reuse of water, contributing to the circular economy and the reduction of environmental impact. From an energy perspective, it promotes the use of renewable energy sources and the optimisation of energy consumption, ensuring a balance between demand and resource availability. The measurement of water footprint is also incorporated, enabling the assessment of environmental impact and the implementation of corrective measures. Finally, environmental awareness is promoted, encouraging communities and organisations to adopt sustainable practices based on data and scientific analysis.

The water–energy nexus is central to this layer. Recent work on digital twin platforms for water distribution (Ghorbani Bam et al. 2025) demonstrates how simulation-based planning can quantify trade-offs between pumping schedules and leakage reduction. Within TRITON, SO-7 (Energy Consumption Optimisation) and

SO-4 (Leak Detection) benefit from joint consideration: hydraulic models embedded in a digital twin can identify pressure regimes that simultaneously minimise energy use and reduce pipe stress, thereby lowering burst frequency. Quintana et al. (2025) corroborate this perspective by reporting that integrated sensing and actuation platforms in smart water systems yield measurable reductions in non-revenue water when energy and leakage objectives are co-optimised. Such simulation capabilities provide the analytical backbone upon which the remaining sustainability indicators—from water reuse (SO-3) to environmental awareness (SO-8)—can be monitored and progressively improved.

5.3 Artificial Intelligence (AI) Layer

As summarised in Table 12, this layer of the framework is designed to optimise resource management through the use of advanced algorithms and IoT sensors. Artificial Intelligence enables the prediction of consumption and demand, facilitating the efficient planning of both water and energy resources. In turn, process automation improves operational efficiency by reducing human intervention and minimising errors. Machine learning allows the system to dynamically adapt to changing conditions, while data-driven decision-making enhances the responsiveness to critical events. Furthermore, intelligent alerts and real-time analytics are integrated to detect anomalies and improve the system's security and performance. Finally, the AI layer incorporates simulation and scenario modelling, enabling the evaluation of potential solutions prior to their implementation and ensuring more informed and strategic decisions.

How much trust can operators place in AI outputs that affect public health? This question runs through the AI layer. Maier et al. (2024) argue that explainable AI (XAI) techniques—such as Shapley-value-based attribution and attention visualisation—are necessary for building operator trust and satisfying regulatory scrutiny in hydrological applications. The EU AI Act (European Parliament and Council of the European Union 2024) reinforces this requirement: AI systems used in the management of critical infrastructure—including water distribution—may fall within the high-risk category, triggering obligations for risk management, human oversight, and performance documentation that IA-4 is designed to facilitate. Within TRITON, IA-4 (Data-driven Decision-making) accordingly requires transparent models whose reasoning can be audited by non-specialist operators. In multi-utility contexts where data sharing raises privacy concerns, federated learning offers a mechanism to train detection models across organisational boundaries without centralising sensitive consumption records; it is not, however, without cost—communication overhead scales with the number of participating nodes, model convergence slows under non-IID data distributions, and the approach remains susceptible to gradient-based poisoning attacks that could compromise the shared model (Liu et al. 2022). Habib et al. (2026) further show that AI-based detection architectures can be embedded directly into network edge devices, reducing latency and reinforcing the cybersecurity posture defined in the CI layer. Experimental evaluations of ML-based intrusion detection in ICS environments (Gauthama Raman et al. 2021) caution that detection performance observed in laboratory conditions does not transfer straightforwardly to operational networks, where data heterogeneity, class imbalance, and adversarial evasion degrade classifier reliability.

From a deployment standpoint, the AI layer must contend with the heterogeneity of field conditions that characterise water distribution networks. Sensor sampling rates, communication bandwidth, and computational resources vary substantially between urban trunk mains and peri-urban branch networks, necessitating a tiered model deployment strategy. Lightweight statistical models (e.g. control charts, exponential smoothing) are suited to edge devices with constrained processing capacity, whilst more computationally demanding architectures—such as convolutional or recurrent neural networks—can be hosted on centralised servers or cloud platforms that aggregate data from multiple districts. Accordingly, TRITON separates model inference from periodic retraining to reduce latency whilst maintaining predictive performance: inference occurs at or near the sensor for time-critical alerts (IA-7), whereas retraining is performed on aggregated datasets incorporating the latest operational and environmental data. This separation aligns with the defence-in-depth principle of the cybersecurity layer, since edge inference reduces the volume of raw data traversing the network and thereby limits the attack surface available to adversaries.

5.4 Relationship Between Layers

To articulate these three fundamental pillars into a cohesive and effective solution, the TRITON Framework has been conceptualised as a systemic architecture aimed at addressing complex challenges in critical technological management scenarios. It places particular emphasis on environmental sustainability, applied artificial intelligence, and cybersecurity. Its foundation lies in the need to integrate multidimensional approaches to build resilient, intelligent, and secure solutions across domains such as water resource management, energy efficiency, and digital infrastructure governance. This integration is graphically represented in Figure 10, which illustrates the relationship between the three strategic axes of the model: Cybersecurity, Sustainability, and Artificial Intelligence.

Each of these axes presents its own essential characteristics. For instance, Cybersecurity is structured around functions such as data protection (CI-1), access control (CI-2), network security (CI-7), and mechanisms such as encryption (CI-5), which ensure the confidentiality, integrity, and availability of information systems. These capabilities are critical in any connected technological environment, and even more so in distributed infrastructures that manage critical resources, as proposed by TRITON. In parallel, the Sustainability axis outlines a set of goals aimed at the efficient and responsible use of natural resources. Features such as water capture and reuse (SO-3), energy consumption optimisation (SO-7), and environmental awareness (SO-8) are positioned as fundamental pillars of an architecture committed to long-term sustainability.

Artificial Intelligence is not presented as an isolated component, but as a transversal axis whose value emerges through its intersections with Cybersecurity and Sustainability. For instance, consumption forecasting (IA-1) and process automation (IA-2) arise at the intersection between AI and Sustainability, where predictive and adaptive capacities are used to optimise resource use. Meanwhile, real-time analytics (IA-5), which belongs to the convergence of all three axes, enables secure and responsive monitoring of environmental and operational parameters. These functions

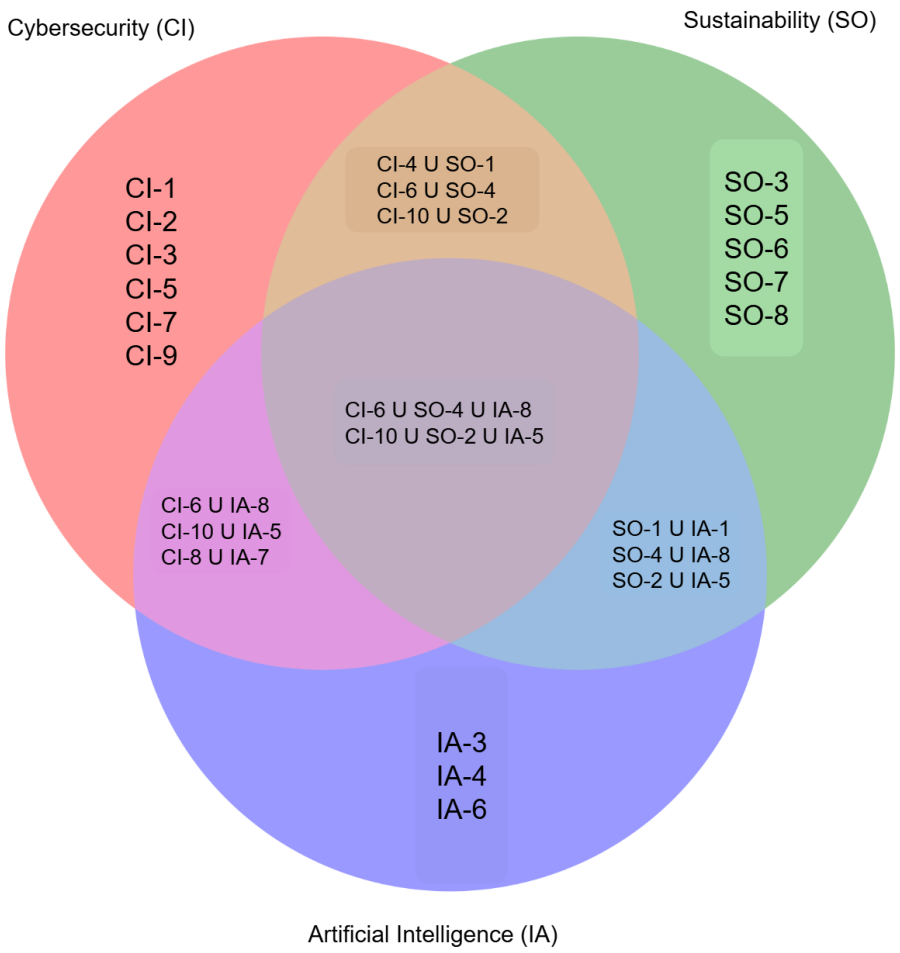


Fig. 10 Relationship Between the TRITON Axes

are not exclusive to AI, but are enabled through the synergy among the TRITON axes. It is precisely in these interconnections where the framework derives its strength, enabling data-driven decision-making that anticipates critical events and enhances system resilience and efficiency. However, the true value of TRITON lies not solely in these individual axes, but in the synergy that arises from their intersections—synergies which are fundamental to ensuring a coherent and effective implementation in real-world environments.

For example, at the intersection between Cybersecurity and Sustainability, relationships emerge such as the one between the security of IoT devices (CI-4) and the efficient use of water (SO-1). Here, sensors monitoring water consumption must be resilient against attacks or tampering that could compromise their accuracy. Likewise, anomaly detection (CI-6) plays a critical role in leak detection (SO-4), allowing the

identification of water losses through anomalous patterns in flow data, thus ensuring both sustainability and infrastructure security. Another key convergence is seen in real-time monitoring (CI-10) combined with quality monitoring (SO-2), enabling safe and continuous supervision of environmental parameters.

At the intersection between Cybersecurity and Artificial Intelligence, the focus shifts to the protection of intelligent systems. Anomaly detection (CI-6) complements pattern recognition (IA-8) in developing AI models capable of identifying threats or unusual behaviours in real time. Furthermore, incident management (CI-8) is enhanced through intelligent alerts (IA-7), which automate responses to critical events, thereby improving the overall resilience of the system. These synergies help consolidate autonomous systems capable of protecting, adapting, and evolving themselves.

The convergence between Sustainability and Artificial Intelligence is primarily centred on dynamic resource optimisation. Consumption forecasting (IA-1) supports strategies for rational water usage (SO-1), while process automation (IA-2) contributes to energy optimisation (SO-7). These relationships enable the development of proactive and adaptive infrastructures that can respond to environmental variability and consumption patterns in real time, generating a positive environmental impact through automated, evidence-based decisions.

Finally, at the centre of the diagram, where the three axes converge, critical functionalities emerge that are key to the success of the framework. The most emblematic case is the relationship between anomaly detection (CI-6), leak detection (SO-4), and pattern recognition (IA-8). This triad allows the construction of systems that identify water losses through AI algorithms trained on historical data, safeguarded by cybersecurity measures. Another example is the combination of real-time monitoring (CI-10), quality monitoring (SO-2), and real-time analytics (IA-5), which facilitates continuous, accurate, and secure environmental surveillance, enabling an immediate response to critical changes in the system.

In practical terms, this layered interdependence translates into a set of operational synergies that directly benefit water network management. The synergies are not merely conceptual: each pairing of features across layers corresponds to a concrete operational capability that can be incrementally deployed, tested, and refined. The following paragraphs describe how the framework translates these inter-layer relationships into actionable functions, with particular attention to deployment considerations in resource-constrained environments.

TRITON operationalises this layered architecture by combining AI-driven analytics with cybersecurity safeguards and sustainability objectives, with particular emphasis on deployability in developing regions.

Regarding security, AI is employed for predictive analysis and real-time anomaly detection (including leaks, contamination, and both physical and cyber intrusions in SCADA systems), while the cybersecurity components—based on standards and regulations—ensure data integrity and confidentiality, as well as the resilience of control infrastructure. This results in reduced water losses due to undetected failures, faster incident response, and protection against threats that could compromise the quality or continuity of supply.

In the field of sustainability, TRITON leverages AI to optimise water distribution, forecast demand patterns, and automate the operation of pumps and valves, thereby minimising energy consumption and water losses (Non-Revenue Water). Additionally, it supports the monitoring of key performance indicators for water and environmental metrics, laying the groundwork for management strategies that may include, for instance, water footprint assessment. The integration of these three axes ensures that improvements in operational efficiency (AI) are carried out securely (Cybersecurity) and with a focus on long-term viability and environmental impact (Sustainability).

It is important to note that specific guidelines for developing regions and the prioritisation of actions may vary significantly. TRITON is conceived as an adaptable and modular framework. In such contexts, the initial priority often lies in securing basic supply and reducing both technical and non-technical water losses. Intelligent alerts (AI-driven to detect leaks, pipe bursts, or anomalous consumption) and foundational cybersecurity for monitoring and control systems (even if basic) tend to provide a more immediate and tangible impact than comprehensive water footprint assessments, which require greater maturity in data collection and management. Smart alerts allow for swift intervention, conserving a scarce resource and enhancing operational efficiency, even with potentially limited infrastructure.

In a resource-constrained scenario, TRITON proposes a phased approach (see Figure 11). The first phase would prioritise the deployment of AI modules focused on critical anomaly detection and basic operational optimisation, along with essential cybersecurity measures to safeguard existing infrastructure and the data it generates.

To operationalise the phased model, Table 16 defines measurable entry and exit criteria for each phase, ensuring that progression is driven by observable performance indicators rather than subjective assessment.

Table 17 positions TRITON relative to established frameworks and recent proposals in the literature. Unlike single-domain standards such as NIST CSF 2.0 (Pascoe et al. 2024), which focus exclusively on cybersecurity governance, or Digital Twin platforms (Ghorbani Bam et al. 2025) oriented towards operational simulation, TRITON integrates cybersecurity, sustainability, and artificial intelligence within a unified architecture. The CISA Zero Trust guidance (Cybersecurity and Infrastructure Security Agency (CISA) 2026) provides robust access-control principles for operational technology but does not address resource optimisation or predictive analytics. Similarly, the smart water system review by Quintana et al. (2025) catalogues technological readiness without proposing an integrated governance model. TRITON addresses these shortcomings by combining the security controls of CI-1 through CI-10 with the sustainability metrics of SO-1 through SO-8 and the analytical capabilities of IA-1 through IA-8 into a single, phased deployment model designed for resource-constrained environments.

For example, ensuring the integrity of sensor data (where sensors are available) and protecting basic control points against unauthorised access would be a crucial first step. While detailed water footprint measurement remains an important sustainability goal, it could be deferred to a later stage—once operations have stabilised, major losses have been reduced, and a more robust and reliable data flow has been established, which earlier TRITON phases would help to build. The framework is designed so that

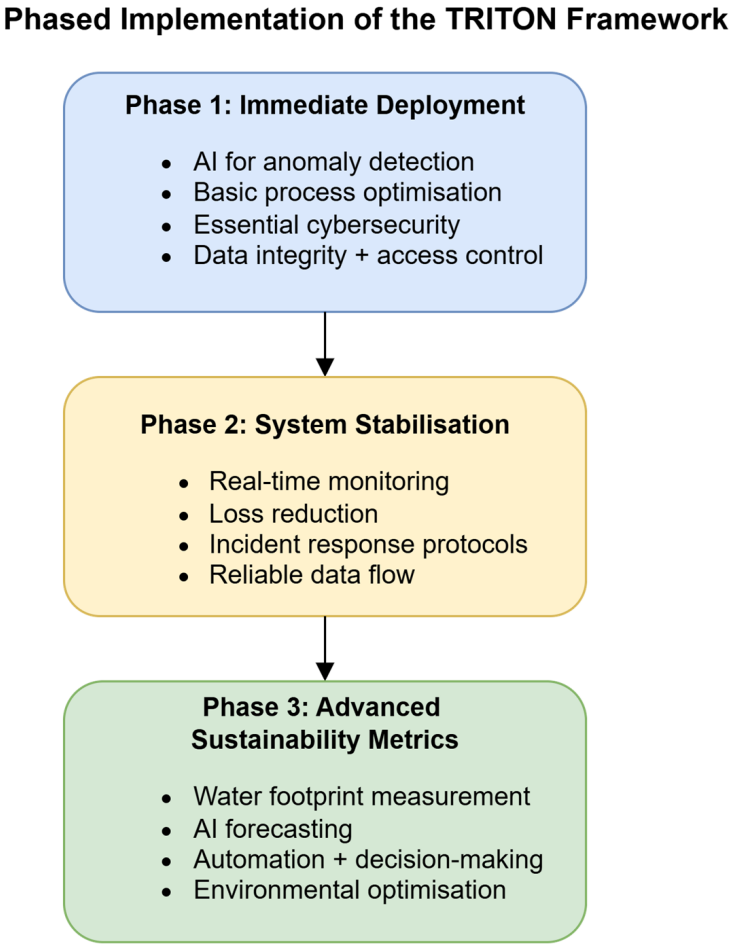


Fig. 11 TRITON phased approach under limited-resource conditions

initial improvements in efficiency and security provide the foundation for more complex sustainability metrics. This pragmatic and adaptive approach is considered essential for TRITON to effectively improve water supply networks in developing regions—starting with the highest-priority needs and progressively building capabilities.

The modular architecture and phased deployment strategy described above are designed to facilitate not only conceptual robustness but also operational deployment and empirical validation in the target domain. Each phase produces observable outputs—reduced non-revenue water, shorter incident response times, or lower energy consumption per cubic metre delivered—that serve as validation milestones and justify continued investment. The modularity of the architecture also permits selective adoption: a utility that has already invested in cybersecurity infrastructure, for example, can integrate the sustainability and AI layers without duplicating existing

Table 16 Phase transition criteria for TRITON deployment. Thresholds are illustrative initial values derived from sector benchmarks and are intended to be calibrated for each utility's operational context.

Phase	Entry Condition	Exit Criterion
1	Basic SCADA/telemetry operational	NRW reduction $\geq$ 10% Kingdom et al. (2006); no confirmed intrusions for 90 consecutive days (validated by external audit or penetration test)
2	Phase 1 criteria met	Real-time monitoring uptime $\geq$ 95%; energy baseline established
3	Phase 2 criteria met	AI model detection performance: precision $\geq$ 0.80, recall $\geq$ 0.85, F1 $\geq$ 0.82 on local validation data; sustainability KPIs tracked

Table 17 Comparative positioning of TRITON against related frameworks

Framework / Standard	Cybersecurity	Sustainability	AI Integration	Water Sector
NIST CSF 2.0 Pascoe et al. (2024)	Comprehensive	Not addressed	Not addressed	Generic CI
CISA ZTA for OT Cybersecurity and Infrastructure Security Agency (CISA) (2026)	Comprehensive	Not addressed	Not addressed	Partial (all CI)
Digital Twin Platforms Ghorbani Bam et al. (2025)	Partial	Partial	Partial	Yes
Smart Water Systems Review Quintana et al. (2025)	Not addressed	Partial	Partial	Yes
AI Detection in WDN Habib et al. (2026)	Partial	Not addressed	Comprehensive	Yes
TRITON (proposed)	By design	By design	By design	Yes

controls. This flexibility is essential for the diverse institutional contexts in which TRITON is intended to operate, ranging from well-resourced European utilities to community-managed systems in developing regions.

5.5 Illustrative Scenario: Leak Detection under Adversarial Conditions

To illustrate how the three TRITON layers interact in practice, consider a leak detection scenario modelled on the conditions documented in the BATADAL dataset Murillo et al. (2022). In Phase 1 of TRITON's deployment, pressure and

flow sensors installed at district metered area boundaries transmit telemetry to an edge gateway. The AI layer (IA-8, Pattern Recognition) applies a lightweight anomaly detector—trained on historical flow profiles—to flag deviations that exceed a configurable threshold. Invariant-based detection methods, which exploit physical constraints inherent to water treatment processes (Yoong et al. 2021), can complement statistical classifiers by providing a model-independent validation layer. When an anomaly is detected, the cybersecurity layer verifies the integrity of the sensor reading: CI-4 (IoT Security) checks the device certificate, CI-5 (Encryption) confirms that the data packet has not been tampered with in transit, and CI-6 (Anomaly Detection) cross-references the alert against known attack signatures to distinguish a genuine leak from a false data injection. Data-driven anomaly detection designed into the communication architecture from inception—rather than retrofitted—has been shown to reduce false-positive rates in cyber-physical production systems (Pinto et al. 2022), a principle that TRITON adopts through its security-by-design approach. If the reading is authenticated, the sustainability layer activates SO-4 (Leak Detection), triggering an intelligent alert (IA-7) to the operations centre and logging the estimated water loss volume for subsequent reporting under SO-1 (Efficient Water Use).

This sequence demonstrates that a single operational event—a pressure drop at a sensor—engages all three layers in a coordinated response. The cybersecurity layer prevents adversarial manipulation of the alert pipeline, the AI layer provides the detection capability, and the sustainability layer ensures that the event is translated into actionable resource management. Without the cross-layer architecture, a utility relying solely on AI-based detection would remain vulnerable to false data injection attacks that could suppress genuine leak alerts or generate spurious ones, whilst a utility with cybersecurity controls but no AI analytics would lack the pattern recognition needed to detect slow-onset leaks that do not trigger simple threshold alarms.

6 Conclusions

This work conducted a systematic review of 40 primary studies—selected from 1 580 records across six academic sources—to determine what processes, frameworks, and models address cybersecurity, sustainability, and artificial intelligence in smart water networks, and to what extent these three domains are treated jointly.

The review confirmed that IoT, Big Data, and cyber-physical systems are actively transforming water network operations, yet the cybersecurity dimension remains critically underrepresented: only 13 of the 467 cybersecurity-related records retrieved met the inclusion criteria. Recurring vulnerabilities—unpatched SCADA firmware, flat OT/IT network topologies, and absent incident response protocols—persist across the reviewed literature. The integration of these domains is highly fragmented: of the 40 studies analysed, none of those included in the final corpus was coded as addressing cybersecurity, sustainability, and AI simultaneously, a finding consistent with recent domain-specific reviews (Habib et al. 2026; Galarza Yallico and Santos López 2024).

To address this gap, the TRITON framework was proposed as a layered architecture designed to integrate ten cybersecurity controls (CI-1 to CI-10), eight sustainability indicators (SO-1 to SO-8), and eight AI capabilities (IA-1 to IA-8). Its phased

deployment model enables utilities to begin with high-impact, low-cost interventions—such as AI-based leak detection and basic access control—before progressing towards advanced capabilities. The lack of a universally adopted, water-specific cybersecurity profile integrating OT security, AI governance, and sustainability metrics further constrains proactive security investment.

The main limitation is that this paper presents TRITON as a conceptual architecture derived from a literature review; it has not yet been empirically validated in an operational WDN environment. A companion paper addresses this limitation by providing the detailed technical specification of each TRITON layer—including algorithms, protocols, and normative alignment with IEC 62443 and ISO/IEC 42001—together with a methodological application to a campus water network and a co-simulation experiment that provides preliminary evidence of cyber-physical event discrimination. Additionally, the review is restricted to English-language publications from 2015–2026, which may exclude relevant work in other languages or from earlier periods. A small number of included studies—notably Alghassab (2021) (energy sector), Kumar et al. (2019) (generic CTI platforms), and Addas et al. (2023) (agricultural robotics)—address adjacent sectors rather than water systems directly; they were retained because their methodological contributions are directly transferable to water network contexts and no water-specific equivalents were identified in the corpus.

Future research should pursue four directions: (i) field deployment of the foundational phase in a resource-constrained water utility to validate inter-module communication and fault tolerance under operational conditions; (ii) integration of emerging security paradigms—Zero Trust Architecture for OT environments (Cybersecurity and Infrastructure Security Agency (CISA) 2026) and alignment with NIST CSF 2.0 (Pascoe et al. 2024)—into TRITON's cybersecurity layer, noting that Zero Trust enforcement in legacy SCADA networks introduces non-trivial latency and may require hardware upgrades at field sites; (iii) incorporation of explainable AI techniques (Maier et al. 2024) to ensure operator trust and regulatory compliance; and (iv) assessment of the computational sustainability of the framework itself, including the energy and water footprint of AI inference at edge and cloud tiers.

Abbreviations. AI, Artificial Intelligence; APT, Advanced Persistent Threat; C2M2, Cybersecurity Capability Maturity Model; CI, Cybersecurity Indicator; CNN, Convolutional Neural Network; CVAE, Conditional Variational Autoencoder; DCS, Distributed Control System; DRL, Deep Reinforcement Learning; FDI, False Data Injection; FISMA, Federal Information Security Management Act; IA, Artificial Intelligence capability identifier in TRITON; ICT, Information and Communication Technology; ICS, Industrial Control System; IDS, Intrusion Detection System; IoT, Internet of Things; IT, Information Technology; ML, Machine Learning; MLP, Multi-Layer Perceptron; NIST, National Institute of Standards and Technology; NIS2, Network and Information Security Directive 2; NRW, Non-Revenue Water; OT, Operational Technology; PLC, Programmable Logic Controller; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RL, Reinforcement Learning; RTU, Remote Terminal Unit; SCADA, Supervisory Control and Data Acquisition; SIEM, Security Information and Event Management; SLR, Systematic Literature Review; SO, Sustainability Indicator; SWaT, Secure Water Treatment

testbed; TRITON, Technology Resilience, cybersecurity and Intelligence for Transformative Optimization of smart water Networks; WADI, Water Distribution testbed; WDN, Water Distribution Network; XAI, Explainable Artificial Intelligence.

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